

(12) **United States Patent**
Guan et al.

(10) **Patent No.:** **US 12,409,983 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **MOBILE COOLING BOX WITH HINGE MODULE**

(71) Applicant: **Dometic Sweden AB**, Solna (SE)

(72) Inventors: **Weixian Guan**, JinWan (CN); **Peng Wang**, JinWan (CN); **Wenmin Tan**, JinWan (CN); **Yuelong Chen**, JinWan (CN)

(73) Assignee: **Dometic Sweden AB**, Solna (SE)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 32 days.

(21) Appl. No.: **18/502,801**

(22) Filed: **Nov. 6, 2023**

(65) **Prior Publication Data**
US 2024/0067414 A1 Feb. 29, 2024

Related U.S. Application Data
(63) Continuation of application No. 17/872,514, filed on Jul. 25, 2022, now Pat. No. 11,827,423, which is a (Continued)

(30) **Foreign Application Priority Data**
Jan. 4, 2019 (DE) 102019200070.3

(51) **Int. Cl.**
B65D 43/16 (2006.01)
B65D 25/28 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **B65D 43/165** (2013.01); **B65D 25/2841** (2013.01); **B65D 43/22** (2013.01); **F25D 3/08** (2013.01)

(58) **Field of Classification Search**
CPC .. B65D 43/165; B65D 43/164; B65D 43/163; B65D 43/169; B65D 43/161;
(Continued)

(56) **References Cited**
U.S. PATENT DOCUMENTS

1,691,468 A 11/1928 Albert
2,012,800 A 8/1935 Allen
(Continued)

FOREIGN PATENT DOCUMENTS

AU 201712770 5/2017
AU 201712777 5/2017
(Continued)

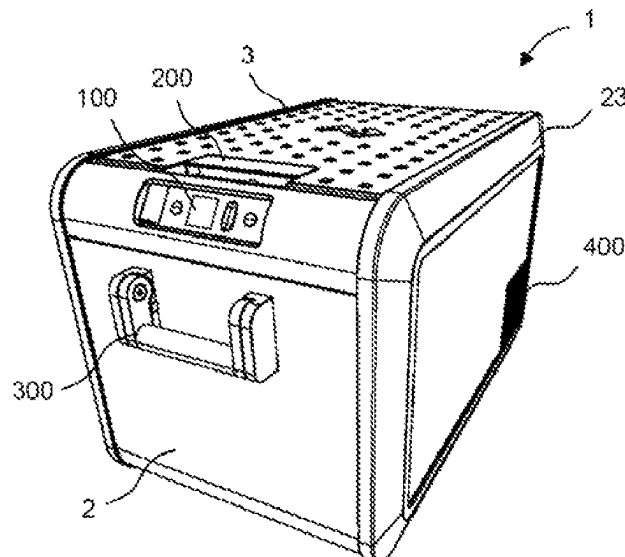
OTHER PUBLICATIONS

US D935,281 S, 11/2021, Thelin et al. (withdrawn)
(Continued)

Primary Examiner — Robert J Hicks
(74) *Attorney, Agent, or Firm* — Dinsmore & Shohl LLP

(57) **ABSTRACT**
A mobile cooling box having a box main body and at least one lid for opening the box and providing access to the inside of the box. The at least one lid is pivotally attached to the box main body of at least two hinge modules. Each hinge module comprises a pin module having a hinge pin with a front end, a rear end, a longitudinal axis about which the lid is pivotable and predominantly a smooth outer surface of a cylindrical shape. Each hinge module further comprises a bearing module having a hinge bearing accommodating the hinge pin. The hinge pin laterally extends with its front end into the hinge bearing. Thus, during pivoting the lid with respect to the box main body an axis of the hinge bearing remains co-linear with the longitudinal axis of the hinge pin.

18 Claims, 17 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/728,657, filed on
Dec. 27, 2019, now Pat. No. 11,396,406.

(51) **Int. Cl.**

B65D 43/22 (2006.01)

F25D 3/08 (2006.01)

(58) **Field of Classification Search**

CPC B65D 43/16; B65D 43/22; B65D 25/2841;
B65D 25/2838; B65D 25/2835; B65D
25/28; A45C 11/20; F25D 3/08; F25D
3/06; F25D 3/02; Y10S 220/9151
USPC 220/592.03, 592.02, 915.2, 915.1, 254.3,
220/254.2, 254.5, 844, 843, 848, 836,
220/810, 830, 827, 826; 62/457.7, 457.2,
62/457.1

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

2,492,601	A	12/1949	Steel	8,910,819	B2	12/2014	Seiders
2,642,728	A	6/1953	Thomaras	D722,474	S	2/2015	Seiders
2,715,817	A	8/1955	Brodheim	D722,475	S	2/2015	Seiders
2,766,901	A	10/1956	Sunko	D732,348	S	6/2015	Seiders et al.
2,926,504	A	3/1960	Hellinger	D732,349	S	6/2015	Seiders et al.
3,035,733	A	5/1962	Knapp	D732,350	S	6/2015	Seiders et al.
3,338,068	A	8/1967	Piker	D732,899	S	6/2015	Seiders et al.
3,796,063	A	3/1974	Wulke et al.	9,137,232	B2	9/2015	Eschbatch et al.
3,938,694	A	2/1976	Archuleta	9,139,352	B2	9/2015	Seiders et al.
3,979,007	A	9/1976	Thornbloom, Jr.	9,163,871	B1	10/2015	Costello
4,162,029	A	7/1979	Esch et al.	9,187,232	B2	11/2015	Seiders
D255,312	S	6/1980	Uyeda	9,220,250	B1	12/2015	Davis, Sr.
D256,587	S	8/1980	Tatsumi	9,232,290	B2	1/2016	Besay
4,328,676	A	5/1982	Reed	D752,347	S	3/2016	Seiders et al.
4,351,165	A	9/1982	Gottsegen et al.	9,282,797	B1	3/2016	Soto
4,468,932	A	9/1984	Bullard	9,320,938	B1	4/2016	Belmore
4,757,920	A	7/1988	Harootian, Jr. et al.	D792,486	S	7/2017	Li
4,802,344	A	2/1989	Livingston et al.	D802,028	S	11/2017	Li
4,866,945	A	9/1989	Bender et al.	D802,029	S	11/2017	Li
4,886,176	A	12/1989	Steakley	D802,630	S	11/2017	Li
5,024,471	A	6/1991	Kahl et al.	9,951,986	B1	4/2018	Turner
5,301,508	A	4/1994	Kahl et al.	9,982,894	B1	5/2018	Philip et al.
5,319,937	A	6/1994	Fritsch et al.	D820,049	S	6/2018	Ahlstrom
D349,007	S	7/1994	Costello	D836,993	S	1/2019	Meda
5,407,218	A	4/1995	Jackson	D836,994	S	1/2019	Meda
5,676,296	A	10/1997	Masters	D844,386	S	4/2019	Ahlstrom
5,860,281	A	1/1999	Coffee et al.	D887,788	S	6/2020	Meda
5,897,435	A	4/1999	Thomas et al.	D888,503	S	6/2020	Meda
6,027,249	A	2/2000	Bielinski	10,717,499	B1	7/2020	Street
6,308,518	B1	10/2001	Hunter	D894,043	S	8/2020	Meda
6,349,845	B1	2/2002	Duncan	D901,986	S	11/2020	Meda
6,354,101	B1	3/2002	Levitin	D920,743	S	6/2021	Meda et al.
6,354,104	B1	3/2002	Feagin	D927,938	S	8/2021	Meda et al.
6,474,097	B2	11/2002	Treppedi et al.	D929,471	S	8/2021	Meda et al.
D497,517	S	10/2004	Reuter	D935,280	S	11/2021	Thelin et al.
D500,057	S	12/2004	Reuter	D954,764	S	6/2022	Meda et al.
6,964,447	B2	11/2005	McNamee	11,396,406	B2	7/2022	Guan
D516,099	S	2/2006	Maruyama	11,415,355	B2	8/2022	Thelin et al.
7,143,601	B1	12/2006	Jimenez	2001/0039807	A1	11/2001	Mogil
D534,771	S	1/2007	Zorn	2002/0095947	A1	7/2002	Treppedi et al.
7,178,673	B1	2/2007	Miller	2003/0019873	A1	1/2003	Nam
7,415,794	B1	8/2008	Thompson	2003/0042266	A1	3/2003	Young et al.
7,617,699	B1	11/2009	Parmely	2003/0106895	A1	6/2003	Kalal
7,913,849	B2	3/2011	Pedrazzi et al.	2004/0178208	A1	9/2004	Leba et al.
D640,102	S	6/2011	Hawkins	2004/0195793	A1	10/2004	Sullivan et al.
8,210,819	B2	7/2012	Wiebe et al.	2005/0274726	A1	12/2005	Boggs et al.
8,312,816	B2	11/2012	Vaccarella	2005/0279123	A1	12/2005	Maldonado et al.
D712,720	S	9/2014	Seiders	2005/0279750	A1	12/2005	Barquist et al.
D712,721	S	9/2014	Seiders	2007/0101754	A1	5/2007	Maldonado
D712,722	S	9/2014	Seiders	2007/0125100	A1	6/2007	Shoenfeld
D712,723	S	9/2014	Seiders	2008/0011759	A1	1/2008	Hsu
D714,125	S	9/2014	Seiders	2009/0126419	A1	5/2009	Yoon et al.
8,827,109	B1	9/2014	Sheehan	2010/0251730	A1	10/2010	Whillock, Sr.
				2011/0226785	A1	9/2011	Sakell
				2013/0068776	A1	3/2013	Patterson
				2013/0334238	A1	12/2013	Goforth
				2015/0008242	A1	1/2015	Kpabar, Jr.
				2015/0021106	A1	1/2015	LaRosa
				2015/0114024	A1	4/2015	Grepper
				2015/0241107	A1	8/2015	Mech
				2015/0322698	A1	11/2015	Seiders
				2015/0353263	A1	12/2015	Seiders et al.
				2016/0101924	A1	4/2016	Mitchell et al.
				2016/0187046	A1	6/2016	Chen et al.
				2017/0023290	A1	1/2017	DeMuth
				2017/0108253	A1	4/2017	Monroy
				2017/0254578	A1	9/2017	Kriesel
				2018/0015938	A1	1/2018	Defrancia
				2018/0100682	A1	4/2018	Nilsen
				2018/0141718	A1	5/2018	Ahlstrom
				2018/0147913	A1	5/2018	Bergin
				2018/0201432	A1	7/2018	Harding
				2018/0306479	A1	10/2018	Vazquez
				2019/0161240	A1	5/2019	Ahlstrom
				2019/0308795	A1	10/2019	Juneau
				2020/0062454	A1	2/2020	Duong et al.
				2020/0216229	A1	7/2020	Weixian
				2020/0217571	A1	7/2020	Weixian
				2020/0217574	A1	7/2020	Thelin

(56)

References Cited**U.S. PATENT DOCUMENTS**

2020/0217575	A1	7/2020	Thelin	
2020/0217576	A1	7/2020	Weixian	
2020/0217578	A1	7/2020	Wang	
2022/0313012	A1 *	10/2022	Ganahl	 B65D 25/32
2022/0355984	A1	11/2022	Guan et al.	

FOREIGN PATENT DOCUMENTS

AU	201712780	5/2017
AU	201712782	5/2017
AU	201712791	5/2017
AU	201712799	5/2017
AU	201712802	5/2017
AU	201712803	5/2017
AU	201712785	7/2017
AU	201716789	11/2017
AU	201716791	11/2017
AU	201717667	1/2018
AU	201717669	1/2018
AU	201717670	1/2018
AU	201717674	1/2018
AU	201717676	1/2018
AU	201717662	2/2018
AU	201816061	1/2019
AU	201910011	2/2019
AU	201913598	8/2019
AU	201913601	8/2019
AU	201913603	8/2019
AU	201913604	8/2019
AU	201913607	8/2019
AU	201913624	8/2019
AU	201913626	8/2019
AU	201913627	8/2019
AU	201913631	8/2019
AU	201913633	8/2019
AU	202012545	6/2020
AU	202012548	6/2020
AU	2019284128	7/2020
AU	2019284129	7/2020
AU	2019284130	7/2020
AU	2019284131	7/2020
AU	2019284133	7/2020
AU	2019284134	7/2020
CN	1888356	A 1/2007
CN	101074057	A 11/2007
CN	101208568	A 6/2008
CN	206488529	U 9/2017
CN	107804580	A 3/2018
CN	207081259	U 3/2018
CN	208312853	U 1/2019
CN	106233083	7/2019
CN	305240920	S 7/2019
CN	110191654	8/2019
CN	209893773	U 1/2020
CN	305553099	1/2020
CN	305751278	5/2020
CN	306079204	9/2020
DE	666767	C 10/1938
DE	20110247	12/2002
DE	202004010081	9/2004
DE	19981898	B3 7/2005
DE	202007001638	5/2007
DE	202009015164	3/2010
DE	102011100722	U1 11/2012
DE	202014000386	U1 4/2014
DE	202013007655	1/2015
DE	102014221784	A1 4/2016
DE	102019200065	1/2019
DE	102019200067	1/2019
DE	102019200068	1/2019
DE	102019200070	1/2019
DE	112017005901	8/2019
DE	102019200063	7/2020
DE	102019200064	7/2020
EA	005230802-001	4/2018

EA	005230802-002	4/2018
EA	001470413-0001-02	1/2019
EA	005948601-0001-30	1/2019
EP	1000577	A1 5/2000
EP	2772704	A1 9/2014
GB	432256	A 7/1935
GB	2419177	A 4/2006
JP	H1194417	A 4/1999
JP	4362188	B2 11/2009
KR	1020050042093	A 5/2005
WO	9936324	A1 7/1999
WO	2014131679	A1 9/2014
WO	2017071654	A1 5/2017
WO	2018095957	5/2018
WO	2018231826	A1 12/2018
WO	2018233611	A1 12/2018
WO	2019206219	A1 12/2018

OTHER PUBLICATIONS

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/711,905 mailed on Sep. 24, 2021.

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on Sep. 24, 2021.

Final Office Action Issued in U.S. Appl. No. 16/728,715 mailed on Oct. 7, 2021.

Non Final Office Action Issued in U.S. Appl. No. 16/728,604 mailed on Oct. 25, 2021.

Notice of Allowance Issued in U.S. Appl. No. 29/785,568 mailed on Nov. 16, 2021.

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/785,568 mailed on Dec. 6, 2021.

United Kingdom Application No. 6157637 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157638 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157639 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157640 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157641 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157642 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157643 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157644 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157645 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157646 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157647 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157648 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157649 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157650 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157651 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157652 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157653 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157654 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157655 titled "Mini Fridge" filed on Aug. 27, 2021.

United Kingdom Application No. 6157656 titled "Mini Fridge" filed on Aug. 27, 2021.

(56)

References Cited**OTHER PUBLICATIONS**

Office Action Issued in German Patent Application No. 102019200070.3 mailed on Aug. 25, 2021.

Office Action Issued in U.S. Appl. No. 16/728,646 mailed on Dec. 21, 2021.

Notice of Allowance Issued in U.S. Appl. No. 16/728,715 mailed on Feb. 2, 2022.

Notice of Allowance Issued in U.S. Appl. No. 29/785,568 mailed on Feb. 8, 2022.

Non Final Office Action Issued in U.S. Appl. No. 16/728,687 mailed on Apr. 15, 2022.

Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on May 2, 2022.

Supplemental Notice of Allowance Issued in U.S. Appl. No. 16/728,604 mailed on May 13, 2022.

Corrected Notice of Allowance Issued in U.S. Appl. No. 16/728,715 mailed on May 16, 2022.

Restriction Requirement Issued in U.S. Appl. No. 16/800,512 mailed on May 24, 2022.

U.S. Appl. No. 17/825,374 titled "Mobile Cooling Box with Ice Maker" filed May 26, 2022.

Design U.S. Appl. No. 29/840,161 titled "Cooler Fender Frame" filed May 26, 2022.

Non-Final Office Action Issued in U.S. Appl. No. 16/728,646 mailed on May 26, 2022.

Design U.S. Appl. No. 29/803,817 titled "Fender Frame, Internal Closure, and Hinge of a Cooler" filed Aug. 16, 2021.

Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on Feb. 7, 2022.

Decision to Grant Issued in German Application No. 102019200070.3 mailed on Mar. 2, 2022.

Corrected Notice of Allowance issued in U.S. Appl. No. 17/886,803 mailed on May 16, 2024.

Corrected Notice of Allowance issued in U.S. Appl. No. 18/464,742 mailed on May 15, 2024.

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/697,071 mailed on Jun. 9, 2021.

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/697,070 mailed on Jun. 9, 2021.

Notice of Allowance Issued in U.S. Appl. No. 29/711,905 mailed on Jul. 12, 2021.

Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on Jul. 12, 2021.

Australian Patent Application No. 2020264299 entitled "Cover for a cooler" filed Nov. 4, 2020.

German Patent Application No. 102020213897.4 titled "Cover for a cooler" filed Nov. 4, 2020.

Amazon.com, ICECO insulated protective cover, pp. 1-2, first date available Oct. 11, 2019. (Year: 2019).

Non Final Office Action Issued in U.S. Appl. No. 16/728,657 mailed on Nov. 15, 2021.

Notice of Allowance Issued in U.S. Appl. No. 16/728,657 mailed on Mar. 24, 2022.

Notice of Allowance Issued in U.S. Appl. No. 16/728,657 mailed on Mar. 9, 2022.

Supplemental Notice of Allowance Issued in U.S. Appl. No. 16/728,657 mailed on May 2, 2022.

Notice of Allowance Issued in U.S. Appl. No. 16/728,657 mailed on Jun. 15, 2022.

Corrected Notice of Allowability Issued in U.S. Appl. No. 16/728,604 mailed on Jul. 8, 2022.

U.S. Appl. No. 17/886,803 titled "Mobile Cooling Box with Handle Module" filed Aug. 12, 2022.

Non-Final Office Action Issued in U.S. Appl. No. 16/728,623 mailed on Oct. 3, 2022.

Non-Final Office Action Issued in U.S. Appl. No. 16/800,512 mailed on Oct. 11, 2022.

Notice of Allowance Issued in U.S. Appl. No. 16/728,646 mailed on Oct. 24, 2022.

Non-Final Office Action issued in U.S. Appl. No. 17/825,374 mailed on Oct. 25, 2022.

Final Office Action Issued in U.S. Appl. No. 16/728,687 mailed on Nov. 28, 2022.

Decision to Grant Issued in German Application No. 102020209896.4 mailed on Jan. 10, 2023.

Final Office Action Issued in U.S. Appl. No. 16/800,512 mailed on Feb. 14, 2023.

Notice of Allowability Issued in U.S. Appl. No. 16/728,646 mailed on Feb. 15, 2023.

Notice of Allowability issued in U.S. Appl. No. 17/825,374 mailed on Mar. 7, 2023.

Corrected Notice of Allowability issued in U.S. Appl. No. 17/825,374 mailed on Mar. 16, 2023.

Non-Final Office Action issued in U.S. Appl. No. 16/728,687 mailed on Mar. 17, 2023.

Corrected Notice of Allowability issued in U.S. Appl. No. 16/728,646 mailed on Apr. 24, 2023.

Notice of Allowability issued in U.S. Appl. No. 17/825,374 mailed on Apr. 24, 2023.

Patent Certificate issued in DE Patent Application No. 102020209896.4 mailed on Apr. 20, 2023.

Final Office Action Issued in U.S. Appl. No. 16/728,623 mailed on May 19, 2023.

Corrected Notice of Allowability issued in U.S. Appl. No. 17/825,374 mailed on Jun. 1, 2023.

Non-Final Office Action issued in U.S. Appl. No. 17/886,803 mailed on May 23, 2023.

Notice of Allowability issued in U.S. Appl. No. 17/825,374 mailed on Jul. 26, 2023.

U.S. Appl. No. 18/464,742 titled "Mobile Cooling Box with Ice Maker" filed Sep. 11, 2023.

Restriction Requirement issued in U.S. Appl. No. 29/803,817 mailed on Aug. 15, 2023.

Corrected Notice of Allowability issued in U.S. Appl. No. 16/728,646 mailed on Aug. 31, 2023.

Office Action issued in CN Application No. 201911066985.7 mailed on Jul. 26, 2023.

Notice of Acceptance issued in AU Application No. 2019284129 mailed on Jan. 8, 2024.

Notice of Allowability issued in U.S. Appl. No. 29/803,817 mailed on Jan. 18, 2024.

Restriction Requirement for Design U.S. Appl. No. 29/697,070 mailed on Dec. 28, 2020.

Notice of Allowance Issued in U.S. Appl. No. 29/697,071 mailed on Dec. 23, 2020.

Examination report mailed in EU Design Application No. 005948601-0001/005948601-0030 on Jan. 9, 2019.

Examination report mailed in DE Patent Application No. 102019200070.3 on Sep. 24, 2019.

Examination report mailed in DE Patent Application No. 102019200064.9 on Sep. 27, 2019.

Examination report mailed in DE Patent Application No. 102019200067.3 on Oct. 8, 2019.

Examination report mailed in DE Patent Application No. 102019200063.0 on Oct. 15, 2019.

Notification to Grant Patent Right for CN Design Application No. 201930331393.8 Mailed on Oct. 29, 2019.

CN Design application No. 201930331393.8 entitled "Cooler" filed on Jun. 25, 2019.

CN Patent Application No. 201911066985.7 entitled "Cover for a Cooler" filed Nov. 4, 2019.

DE Design Application No. 102019200067.3 entitled "Mobile Cooling Box With Ice Maker" filed Jan. 4, 2019.

DE Patent Application No. 102019200063.0 entitled "Mobile Cooling Box with User Interface Module" filed Jan. 4, 2019.

DE Patent Application No. 102019200064.9 entitled "Mobile Cooling Box with Latch Handle Opening" filed Jan. 4, 2019.

DE Patent Application No. 102019200070.3 entitled "Mobile Cooling Box With Hinge Module" filed Jan. 4, 2019.

European Design Application No. 005948601 "Cooler" filed Jan. 4, 2019.

(56)

References Cited**OTHER PUBLICATIONS**

Design U.S. Appl. No. 29/666,264 entitled "Cooler" filed Oct. 11, 2018.

Design U.S. Appl. No. 29/650,910, filed Jun. 11, 2018 titled "Zipper Pull".

AU Patent Application No. 202012545 entitled "Cover for Cooler" filed May 1, 2020.

AU Patent Application No. 202012548 entitled "Cover for Cooler" filed Apr. 30, 2020.

European Design Application No. 007847777-0002 "CFX3 Protective covers" filed Apr. 30, 2020.

Decision to Grant mailed in DE Patent Application No. 102019200064.9 on Jul. 31, 2020.

Decision to Grant mailed in DE Patent Application No. 102019200063.0 on Aug. 4, 2020.

Restriction Requirement for Design U.S. Appl. No. 29/697,069 mailed on Sep. 24, 2020.

AU Patent Application No. 2019284133 entitled "Mobile cooling box with ice maker" filed Dec. 30, 2019.

AU Patent Application No. 2019284131 entitled "Mobile cooling box with hinge moduler" filed Dec. 30, 2019.

AU Patent Application No. 2019284128 entitled "Mobile cooling box with user interface moduler" filed Dec. 30, 2019.

AU Patent Application No. 2019284134 entitled "Mobile Cooling Box with Latch Handle Opening" filed Dec. 30, 2019.

Notice of Allowance Issued in U.S. Appl. No. 29/697,069 mailed on Feb. 1, 2021.

Decision to Grant mailed in DE Patent Application No. 1020192000681 on Oct. 15, 2020.

Examination report mailed in DE Application No. 102019200065.7 on Sep. 30, 2019.

Examination report mailed in DE Application No. 102019200068.1 on Oct. 8, 2019.

DE Patent application 102020209896.4 entitled "Handle Module" filed on Aug. 5, 2020.

Decision to Grant mailed in DE Patent Application No. 102019200065.7 on Aug. 4, 2020.

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/697,069 mailed on Feb. 26, 2021.

Corrected Notice of Allowance Issued in U.S. Appl. No. 16/728,715 mailed on Feb. 23, 2022.

Supplemental Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on Mar. 10, 2022.

Notice of Allowance Issued in U.S. Appl. No. 29/785,568 mailed on Mar. 9, 2022.

Corrected Notice of Allowance Issued in U.S. Appl. No. 16/728,715 mailed on Mar. 9, 2022.

Notice of Allowance Issued in U.S. Appl. No. 16/728,604 mailed on Apr. 13, 2022.

Notice of Allowance Issued in U.S. Appl. No. 29/697,071 mailed on Apr. 12, 2021.

United States Patent Office, Notice of Allowance in U.S. Appl. No. 29/697,070 dated Apr. 28, 2021.

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/697,070 mailed on May 19, 2021.

Design U.S. Appl. No. 29/785,568, filed May 26, 2021 titled "Cooler Fender Frame".

Non Final Office Action Issued in U.S. Appl. No. 16/728,715 mailed on May 19, 2021.

Corrected Notice of Allowance for U.S. Appl. No. 29/697,070 dated Jun. 3, 2021.

Corrected Notice of Allowance for U.S. Appl. No. 29/697,071 dated Jun. 3, 2021.

Examination Report Issued for DE Application No. 102020209896.4 mailed on Apr. 19, 2021.

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/711,905 mailed on Aug. 11, 2021.

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on Aug. 11, 2021.

Restriction Requirement Issued in U.S. Appl. No. 16/728,623 mailed on Aug. 30, 2021.

Design U.S. Appl. No. 29/585,279, filed Nov. 22, 2016 titled Latch.

Design U.S. Appl. No. 29/605,186, filed May 24, 2017 titled Refrigerating Apparatus.

Design U.S. Appl. No. 29/650,906, filed Jun. 11, 2018 titled Soft Bag Cooler.

Design U.S. Appl. No. 29/670,885, filed Nov. 20, 2018 titled Cooler.

Design U.S. Appl. No. 29/670,888, filed Nov. 20, 2018 titled Cooler.

Design U.S. Appl. No. 29/697,069, filed Jul. 3, 2019 titled Cooler.

Design U.S. Appl. No. 29/697,070, filed Jul. 3, 2019 titled Cooler.

Design U.S. Appl. No. 29/697,071, filed Jul. 3, 2019 titled Cooler.

Design U.S. Appl. No. 29/708,155, filed Oct. 3, 2019 titled Latch.

Design U.S. Appl. No. 29/708,156, filed Oct. 3, 2019 titled Latch.

Design U.S. Appl. No. 29/711,905, filed Nov. 4, 2019 titled Cover for a Cooler.

Design U.S. Appl. No. 29/711,906, filed Nov. 4, 2019 titled Cover for a Cooler.

U.S. Appl. No. 16/728,646, filed Dec. 27, 2019 titled Mobile Cooling Box with Air Vents.

U.S. Appl. No. 16/728,604, filed Dec. 27, 2019 titled Mobile Cooling Box with Handle Module.

U.S. Appl. No. 16/728,715, filed Dec. 27, 2019 titled Mobile Cooling Box with Ice Maker.

U.S. Appl. No. 16/728,687, filed Dec. 27, 2019 titled Mobile Cooling Box with Latch Handle.

U.S. Appl. No. 16/728,623, filed Dec. 27, 2019 titled Mobile Cooling Box with User Interface Module.

U.S. Appl. No. 16/800,512, filed Feb. 25, 2020 titled Cover for a Cooler.

Design U.S. Appl. No. 29/743,180 filed Jul. 20, 2020 titled Cooler.

Dometic Product Catalog—Coolers; 2015.

Dometic Product Catalog—Coolers; 2016.

Non-Final Office Action issued in U.S. Appl. No. 16/728,623 mailed on Mar. 14, 2024.

Corrected Notice of Allowance issued in U.S. Appl. No. 29/840,161 mailed on Mar. 14, 2024.

Notice of Allowance issued in U.S. Appl. No. 17/886,803 mailed on Mar. 20, 2024.

U.S. Appl. No. 18/751,707 titled "Mobile Cooling Box With Handle Module" filed Jun. 24, 2024.

Office Action issued in CN Application No. 201911066985.7 mailed on Mar. 23, 2024.

U.S. Appl. No. 16/728,657 entitled "Mobile Cooling Box with Hinge Module" filed Dec. 27, 2019.

U.S. Appl. No. 17/872,514 titled "Mobile Cooling Box with Ice Maker" filed Jul. 25, 2022.

Ex-Parte Quayle Action issued in U.S. Appl. No. 17/872,514 mailed on May 5, 2023.

Notice of Allowability issued in U.S. Appl. No. 17/872,514 mailed on Jul. 21, 2023.

Supplemental Notice of Allowability issued in U.S. Appl. No. 17/872,514 mailed on Aug. 30, 2023.

Supplemental Notice of Allowability issued in U.S. Appl. No. 17/872,514 mailed on Sep. 13, 2023.

Final Office Action issued in U.S. Appl. No. 17/886,803 mailed on Oct. 10, 2023.

Corrected Notice of Allowability issued in U.S. Appl. No. 16/728,646 mailed on Oct. 4, 2023.

Supplemental Notice of Allowability issued in U.S. Appl. No. 17/872,514 mailed on Oct. 10, 2023.

Notice of Allowability issued in U.S. Appl. No. 16/728,687 mailed on Oct. 10, 2023.

Examination Report issued for AU Application No. 2019284129 mailed on Dec. 5, 2023.

Notice of Allowance issued in U.S. Appl. No. 29/803,817 mailed on Dec. 6, 2023.

Corrected Notice of Allowability issued in U.S. Appl. No. 16/728,646 mailed on Oct. 25, 2023.

AU Application No. 2024200685 titled "Mobile cooking box with handle module" filed on Feb. 5, 2024.

U.S. Appl. No. 18/420,358 titled "Mobile Cooling Box with Latch Handle Opening" filed Jan. 23, 2024.

(56)

References Cited

OTHER PUBLICATIONS

Notice of Allowance issued in U.S. Appl. No. 29/803,817 mailed on Feb. 8, 2024.

Corrected Notice of Allowability issued in U.S. Appl. No. 29/840,161 mailed on Feb. 16, 2024.

Notice of Allowance issued in U.S. Appl. No. 18/464,742 mailed on Apr. 24, 2024.

Corrected Notice of Allowance issued in U.S. Appl. No. 29/840,161 mailed on May 1, 2024.

Corrected Notice of Allowance issued in U.S. Appl. No. 29/840,161 mailed on Apr. 18, 2024.

U.S. Appl. No. 29/933,756 titled "Internal Closure and Hinge of a Cooler" filed Mar. 21, 2024.

AU Patent Application No. 2025202568 entitled "Mobile cooling box with hinge moduler" filed Apr. 10, 2025.

* cited by examiner

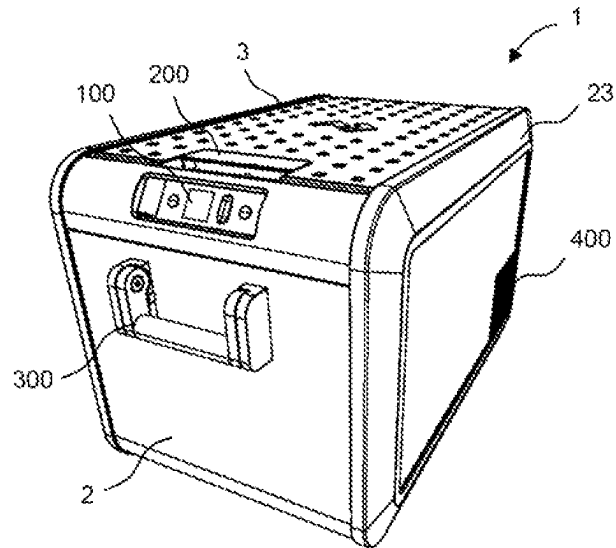


Fig. 1

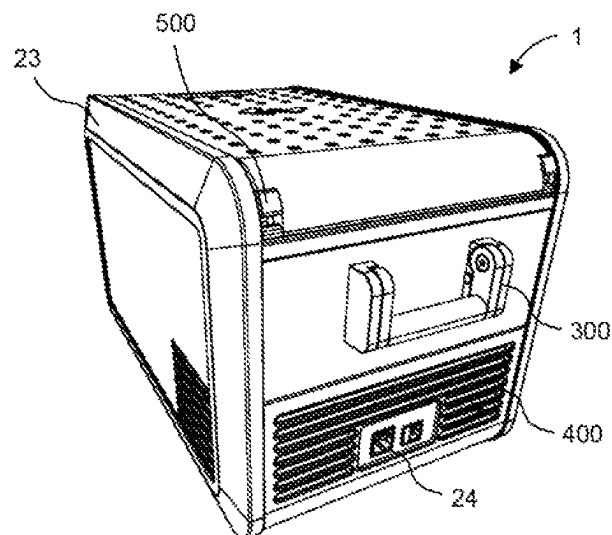


FIG. 2

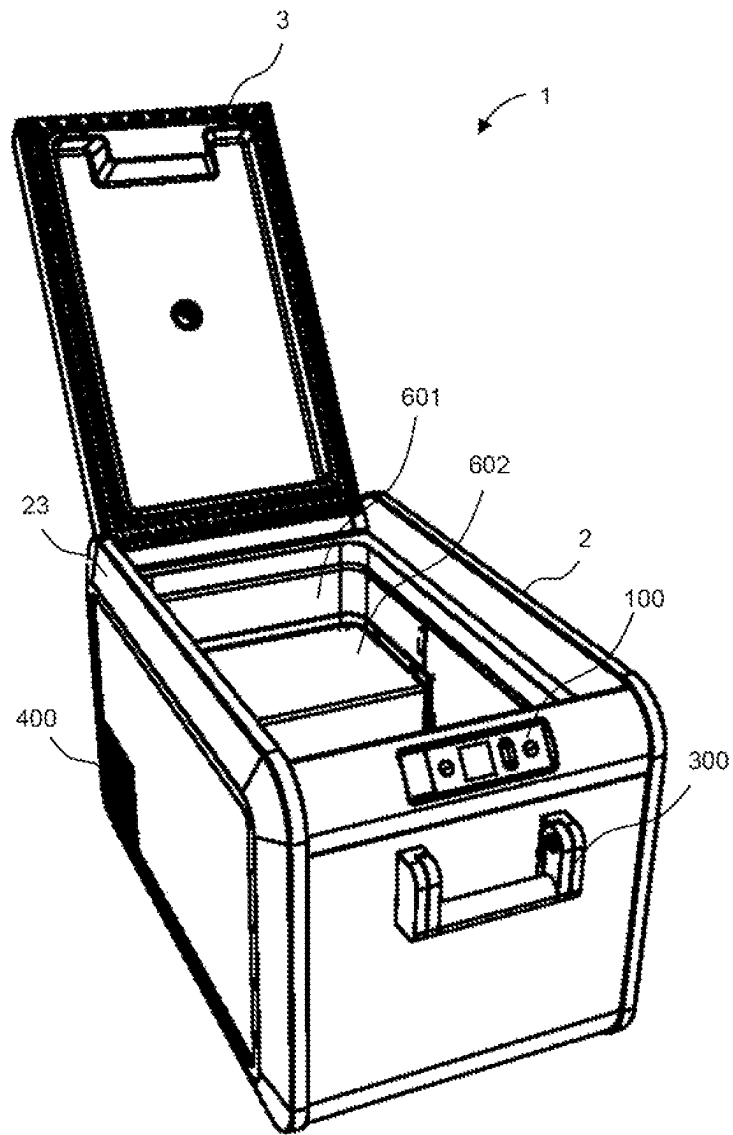


FIG. 3

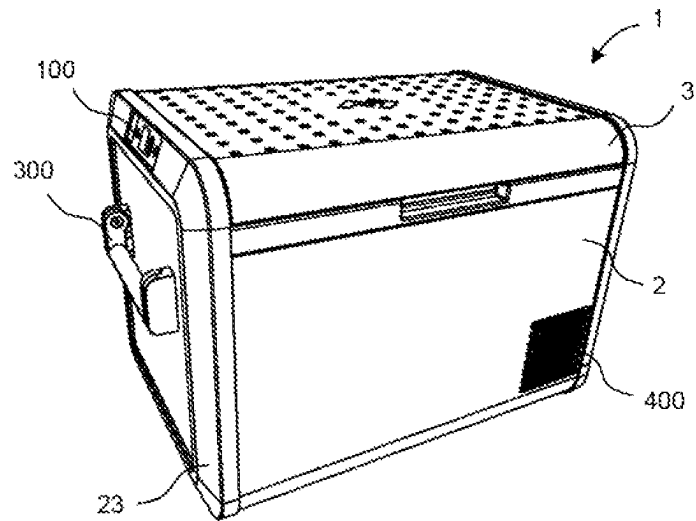


FIG. 4

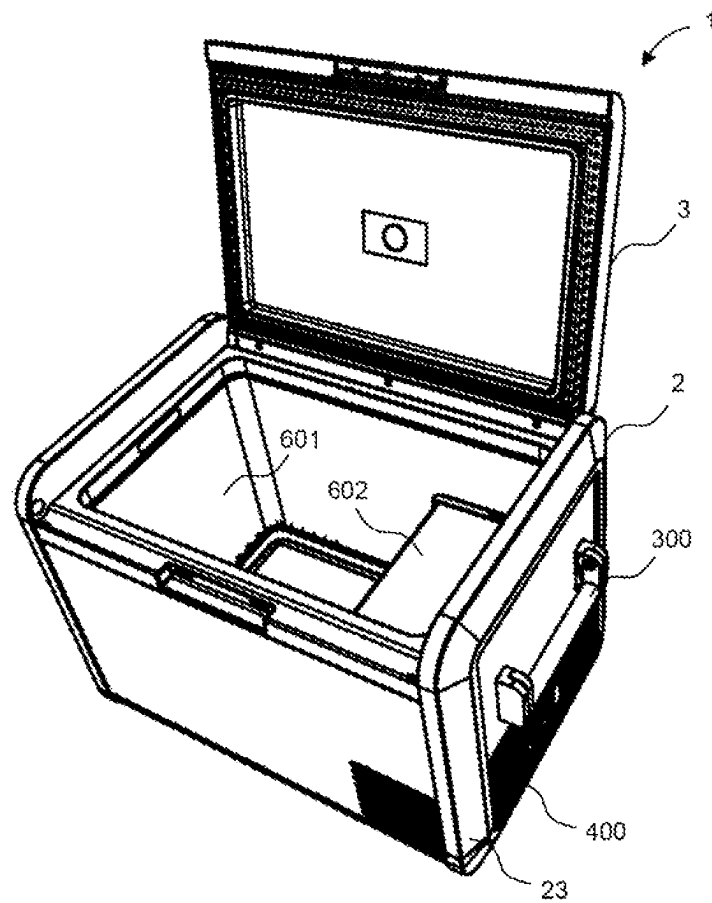


FIG. 5

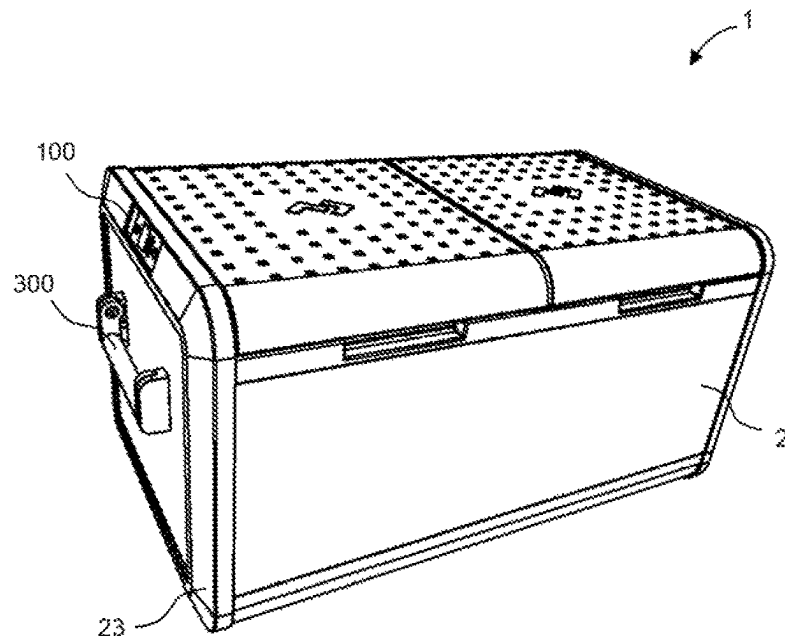


FIG. 6

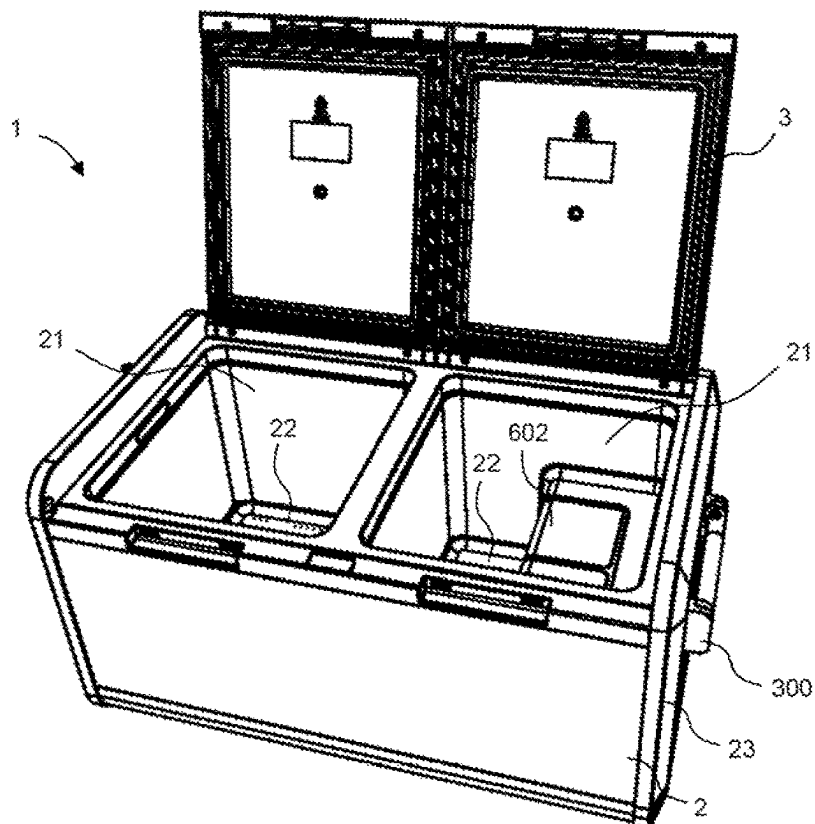


FIG. 7

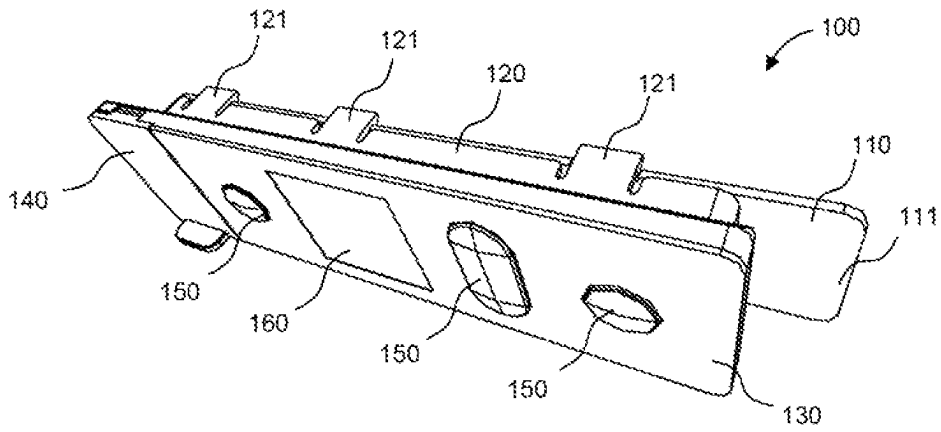


FIG. 8

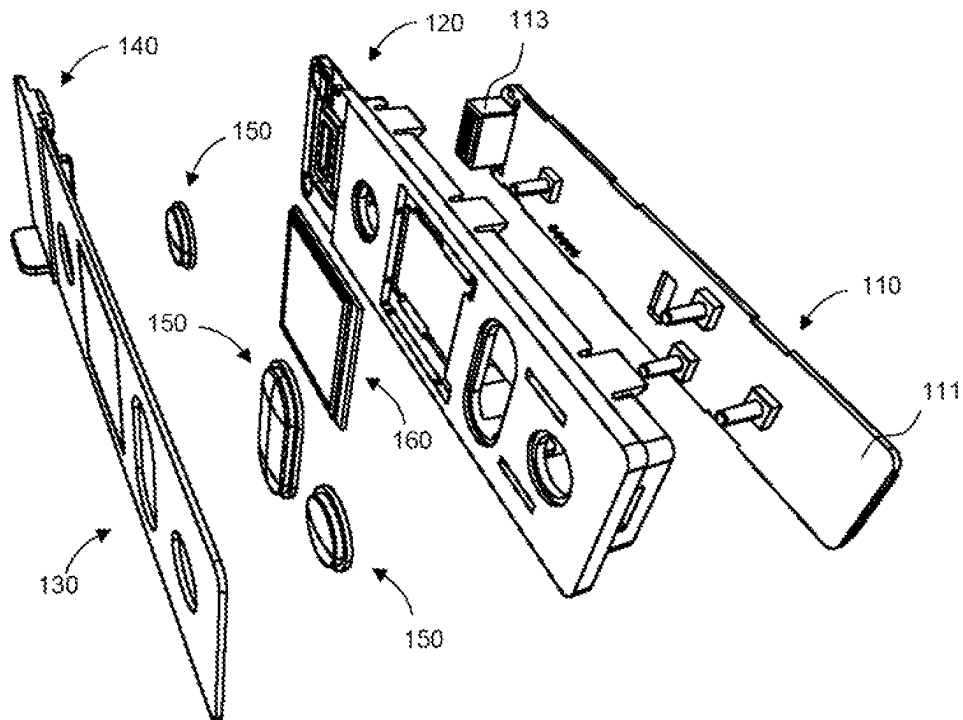


FIG. 9

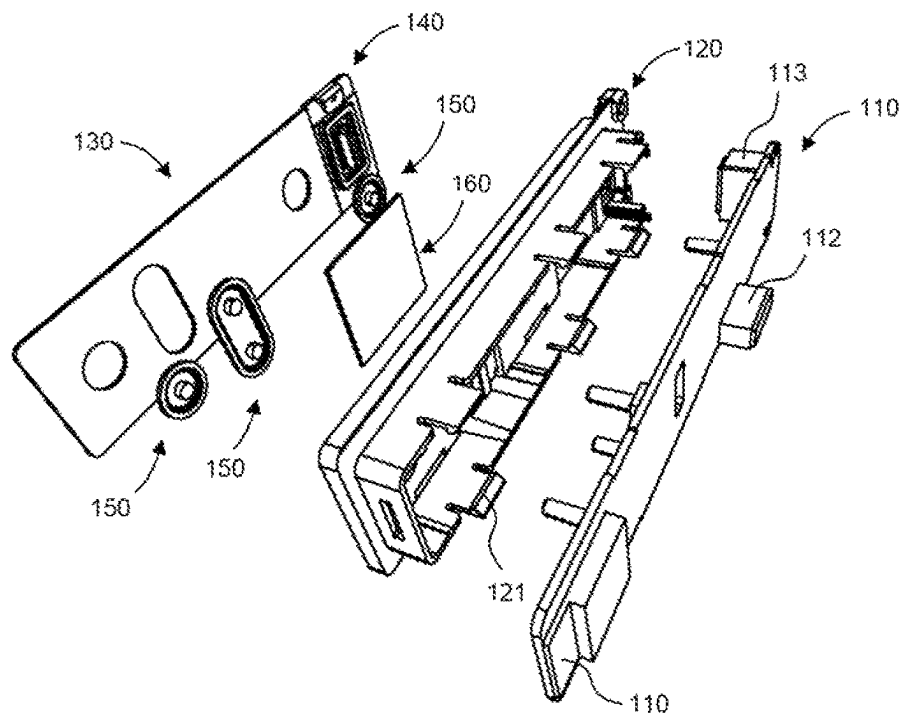


FIG. 10

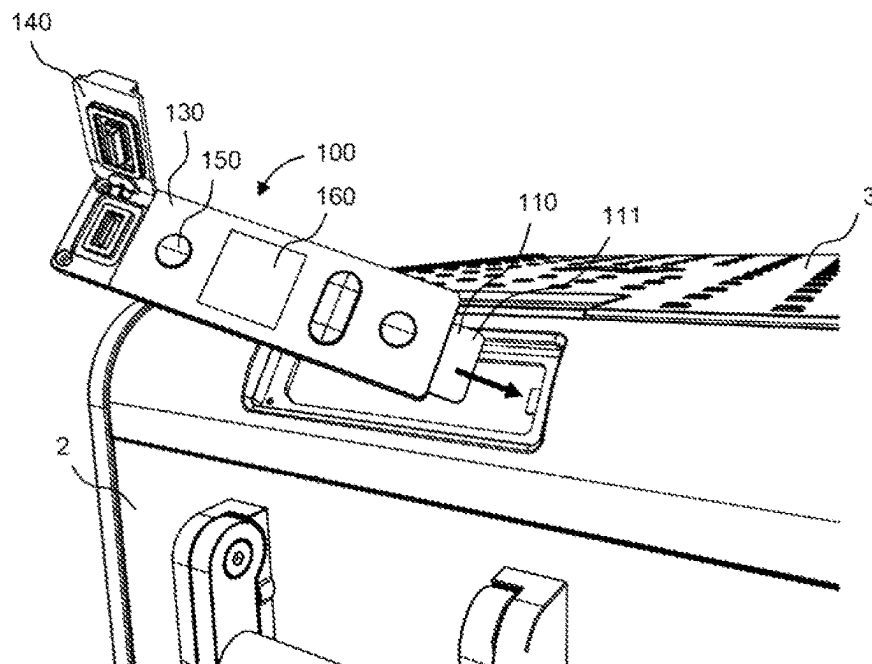


FIG. 11

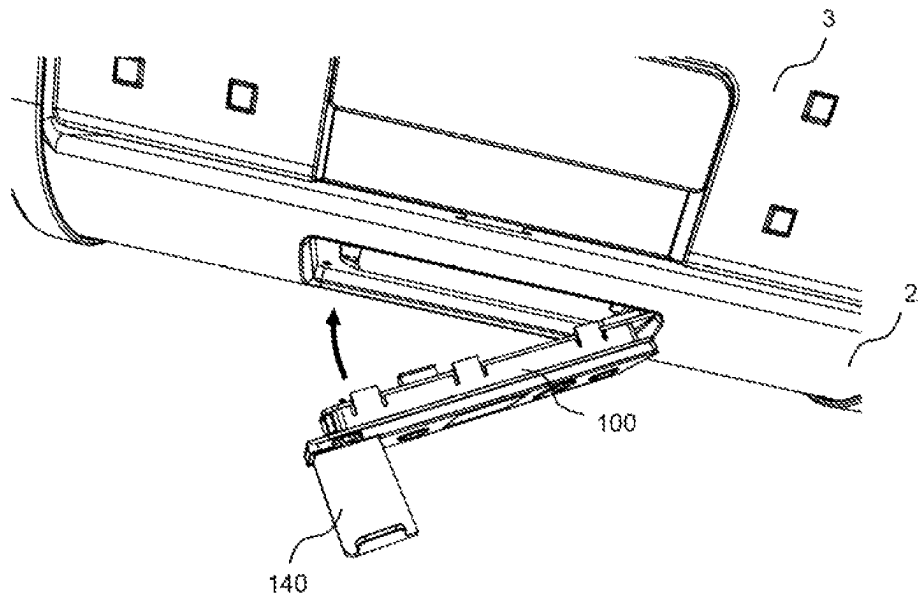


FIG. 12

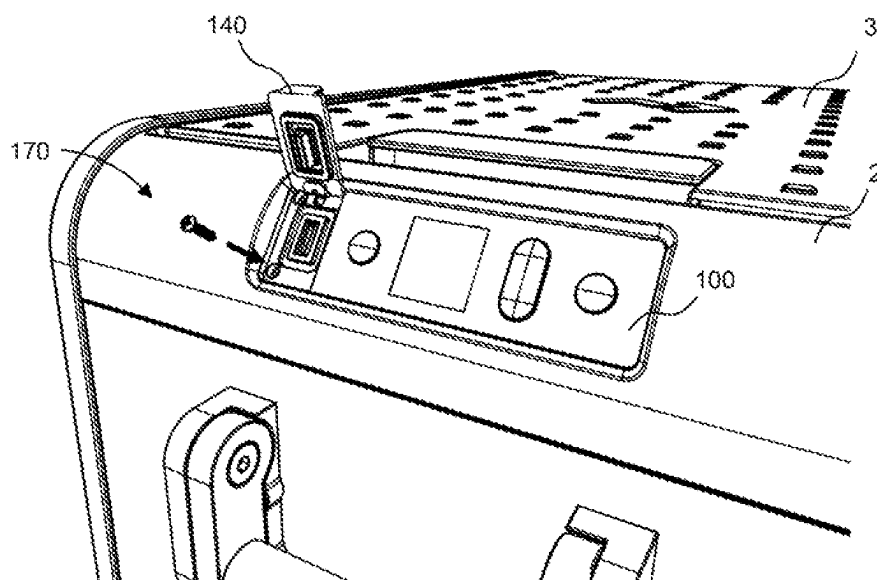


FIG. 13

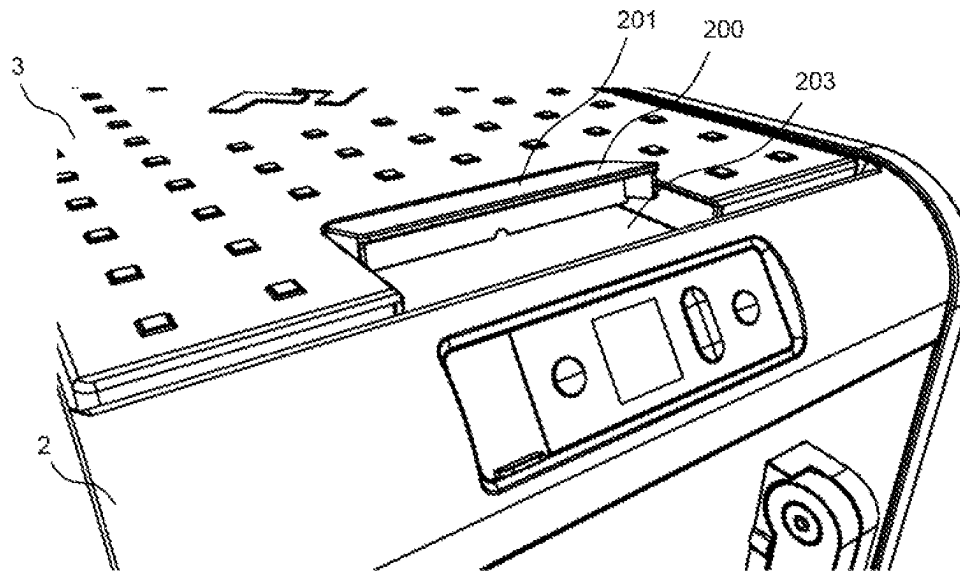


FIG. 14

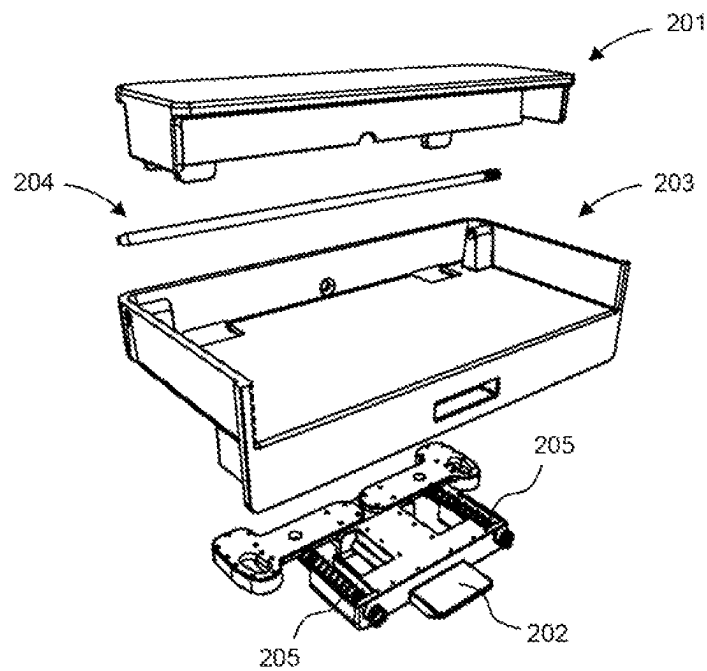


FIG. 15

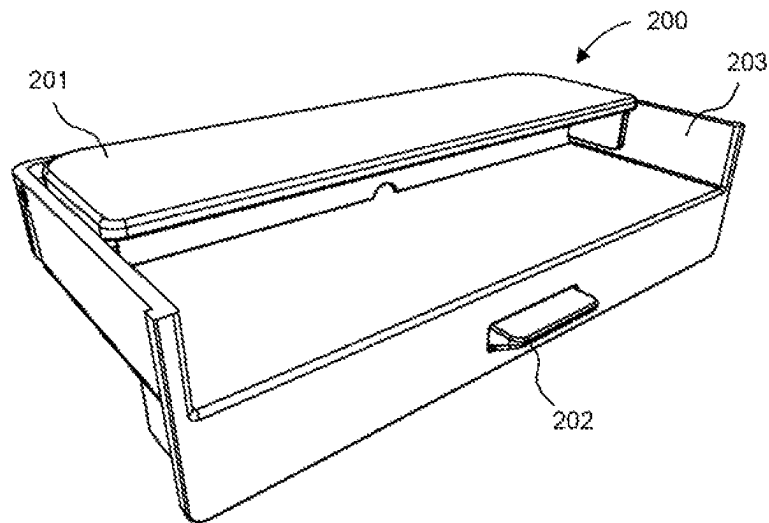


FIG. 16

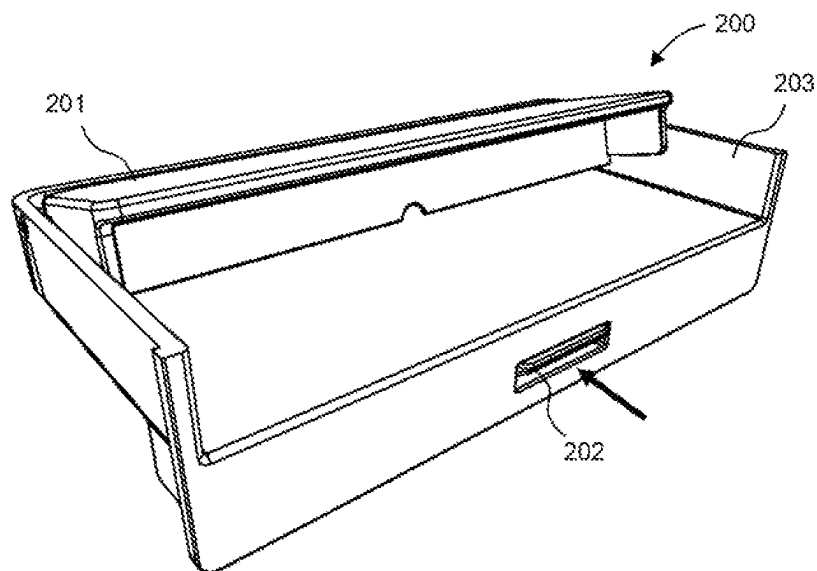


FIG. 17

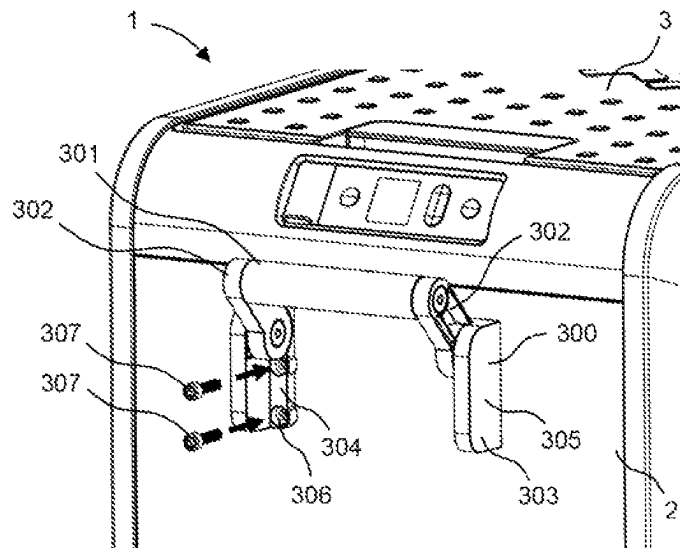


FIG. 18

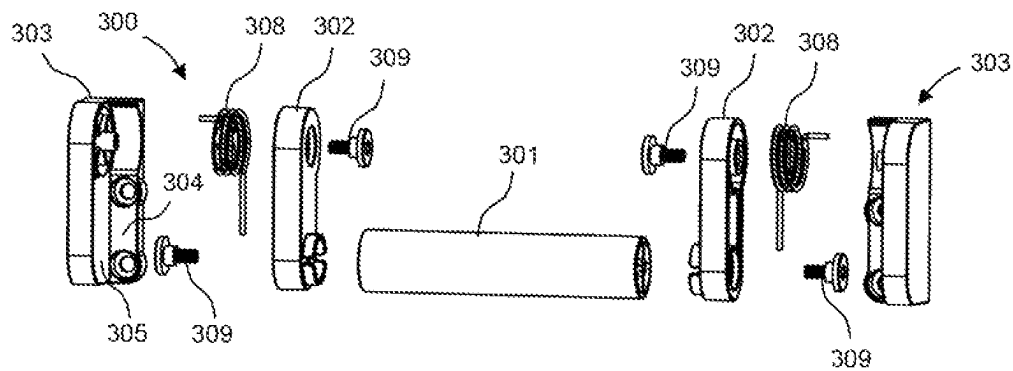


FIG. 19

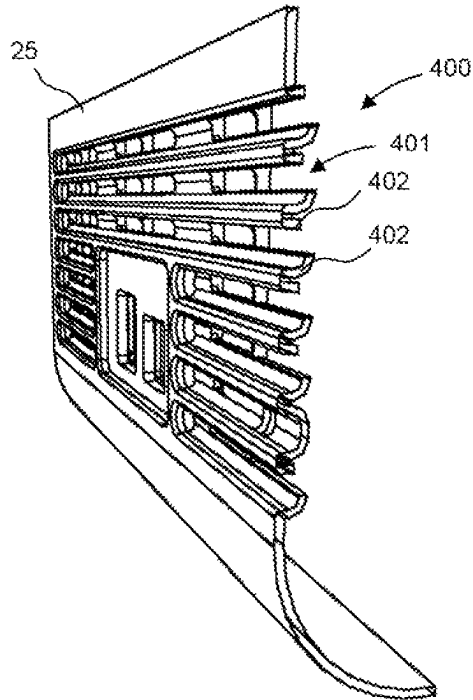


FIG. 20

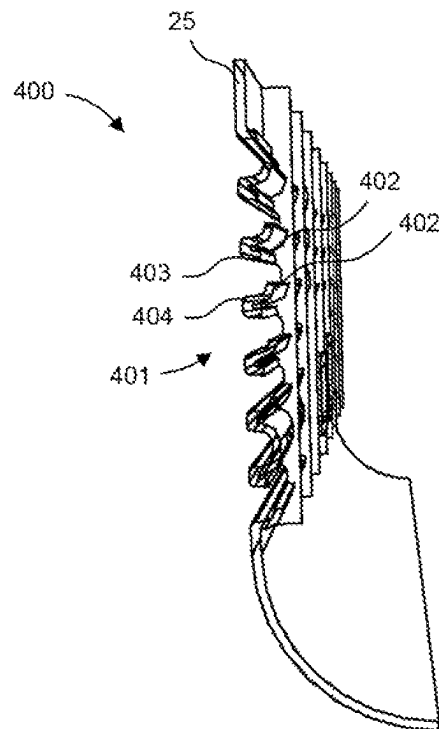


FIG. 21

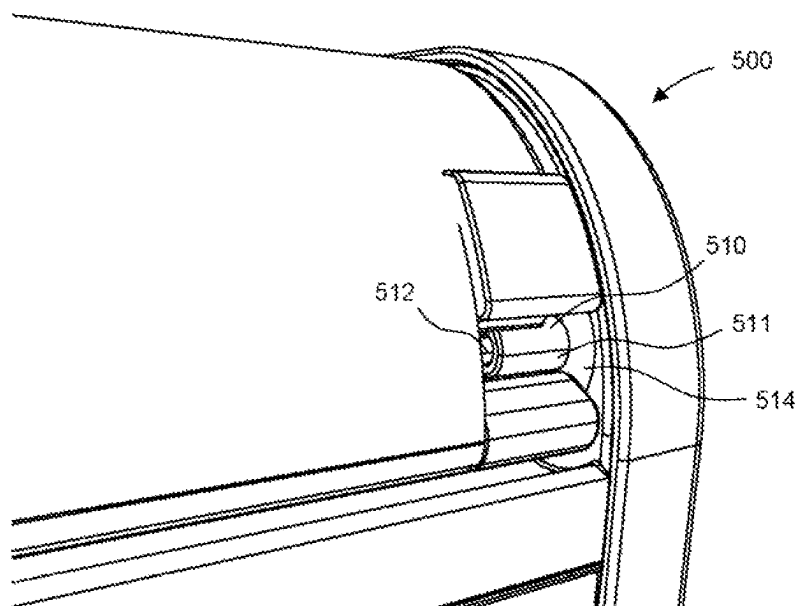


FIG. 22

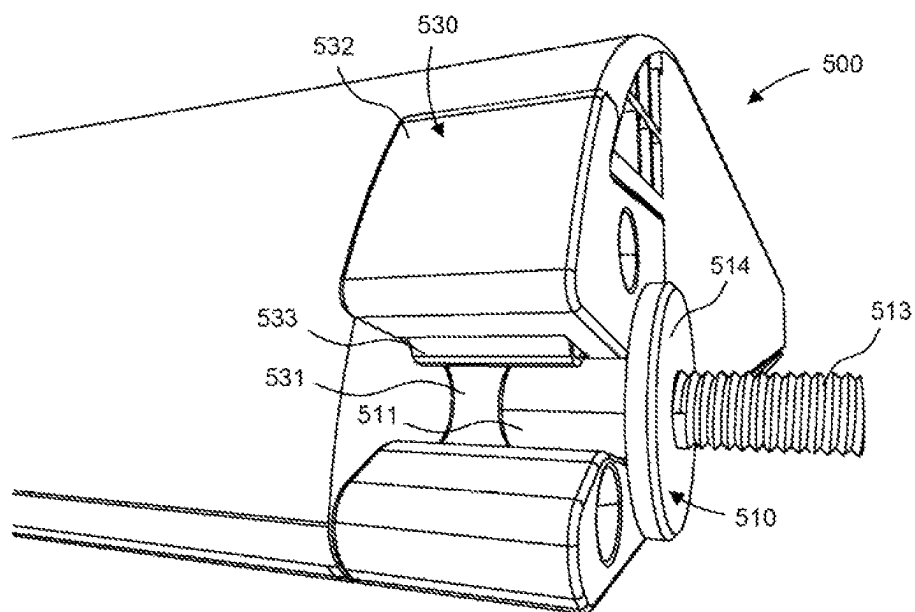
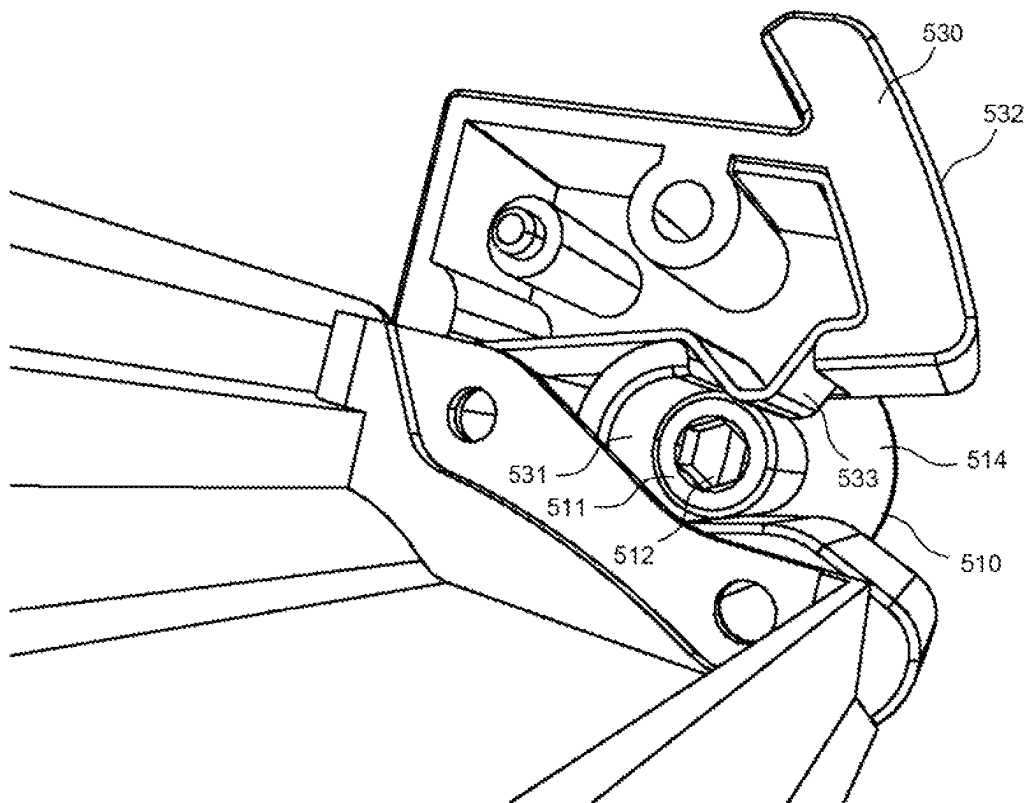


FIG. 23

**FIG. 24**

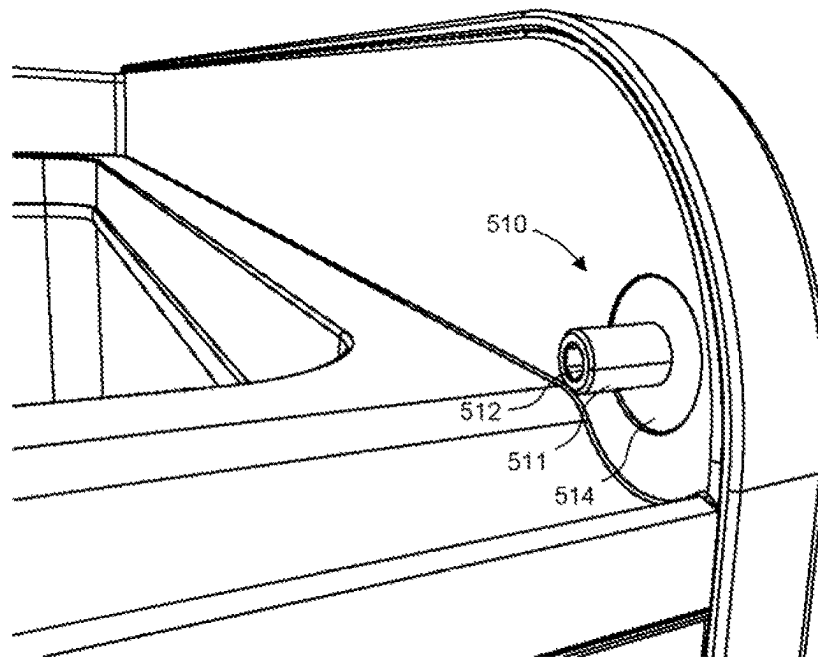


FIG. 25

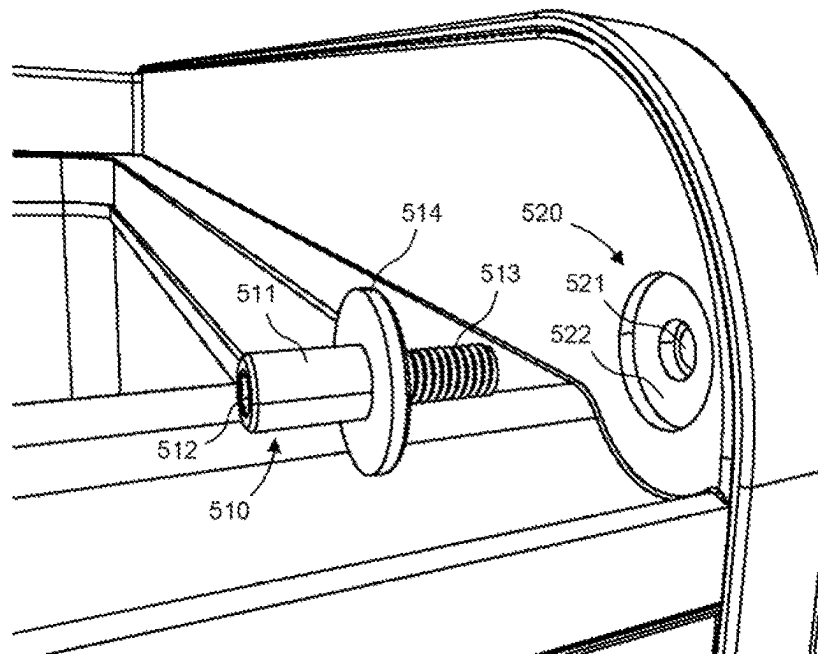


FIG. 26

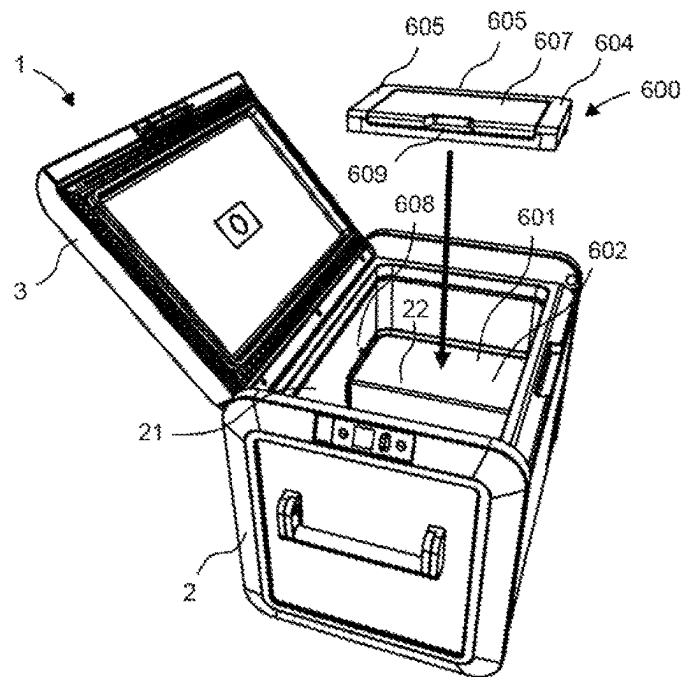


FIG. 27

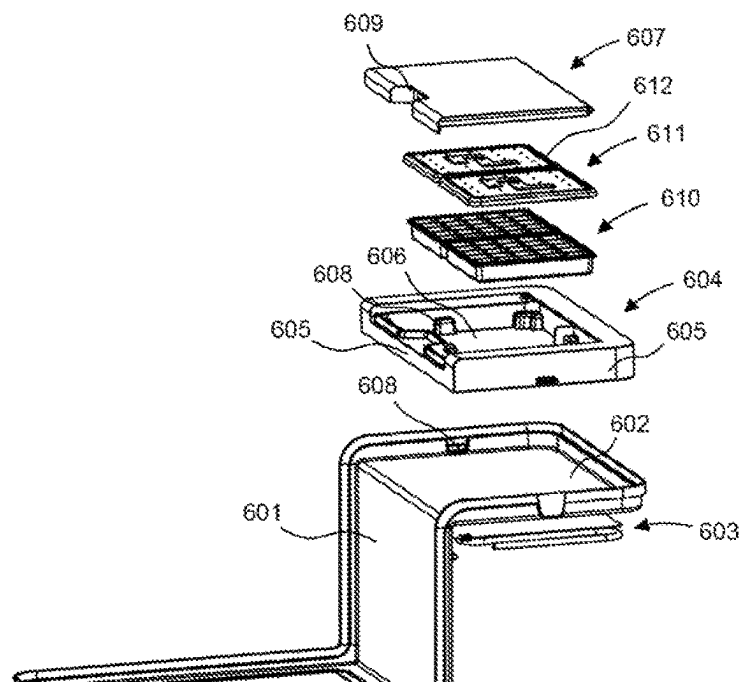


FIG. 28

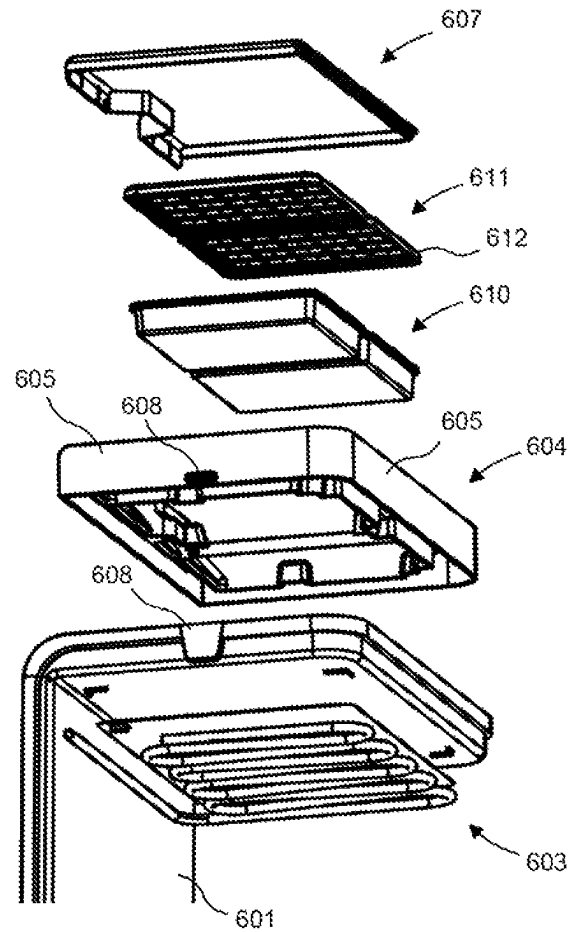


FIG. 29

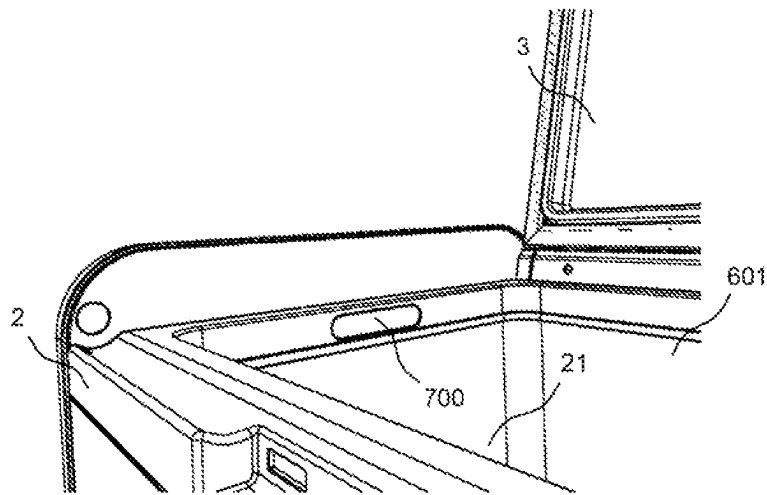


FIG. 30

MOBILE COOLING BOX WITH HINGE MODULE

CLAIM TO PRIORITY

This continuation patent application claims priority to and benefit of, under 35 U.S.C. § 120, U.S. patent application Ser. No. 17/872,514, filed Jul. 25, 2022, which claims priority to U.S. patent application Ser. No. 16/728,657, now issued as U.S. Pat. No. 11,396,406, filed Dec. 27, 2019, which claims priority to German patent application DE 102019200070.3, filed Jan. 4, 2019, all of which is incorporated by reference herein.

BACKGROUND

The present embodiments relate to an improved mobile cooling box with a lid that is pivotally openable by means of a hinge module to enable access to the inside of the mobile cooling box.

Mobile cooling boxes are well-known from the state of the art. Usually, a mobile cooling box comprises a thermo-isolated housing for storing the goods to be stored and cooled inside and a lid for allowing access to the inside to put into or remove the goods from the mobile cooling box. It is within the nature of mobile cooling boxes that such boxes are not stationary but are movable by the user. Typically, such mobile cooling boxes are used for any kind of non-stationary use, like for example during outdoor activities, camping, yachting or the like to store and cool goods like food, drinks or even medical products, etc. for a certain period of time.

Known mobile cooling boxes that are openable by pivoting open a lid usually have plastic hinges. Hereby, the plastic of the lid usually forms some kind of pins that engage with respective counterparts like holes at the box main body. In products of higher quality, the pins are sometimes made of metal. Fixation of the pins, especially the of the ones made of metal requires several additional parts like screws, etc. and, thus, requires additional effort and knowledge when assembling the mobile cooling box. Moreover, the pins have to be adjusted correctly during assembling process in order to achieve a correct orientation of the pins with regard to the receiving counterparts so that a smooth and correct function of the lid is guaranteed. Moreover, it is necessary to provide further force guiding and receiving means to enable the pins to withstand the forces occurring during movement of the probably quite heavy lid, i.e. during opening and closing of the lid by the user. This raises the assembling effort and makes assembling process time consuming and, thus, expensive. Moreover, by providing the plenty of parts the overall weight of the mobile cooling box is increased.

Hence, although there are possibilities available in the prior art for a working hinge arrangement for the lid of a mobile cooling box which basically fulfill the requirements for such devices, taking the above drawbacks of known hinge arrangements of mobile cooling boxes into account there is indeed room for improvements in this regard.

The information included in this Background section of the specification, including any references cited herein and any description or discussion thereof, is included for technical reference purposes only and is not to be regarded subject matter by which the scope of the invention is to be bound.

SUMMARY

The present application discloses one or more of the features recited in the appended claims and/or the following features which alone or in any combination, may comprise patentable subject matter.

The present embodiments, therefore, provide a mobile cooling box with a cost-efficient and reliable and robust hinge construction for the lid of a mobile cooling box.

In order to solve the posed problem, the present embodiments provide a mobile cooling box having a box main body and at least one lid for opening the box and providing access, for example, from above, to the inside of the box. The at least one lid may be pivotally attached to the box main body by at least two hinge modules. Each hinge module has a pin module. The pin module has a hinge pin with a front end, a rear end, a longitudinal axis about which the lid is pivotable and predominantly a smooth outer surface of a cylindrical shape, and a bearing module having a hinge bearing accommodating the hinge pin. The hinge pin laterally extends with its front end into the hinge bearing. Hence, during pivoting the lid with respect to the box main body an axis of the hinge bearing remains co-linear with the longitudinal axis of the hinge pin.

According to the one embodiment, for the mounting of the pin module, the pin module further comprises an engaging portion extending from the rear end of the hinge pin and having a longitudinal axis being co-linear to that of the hinge pin. Thus, the pin module is mountable by the engaging portion.

Thereby, the hinge pin is easily mountable without the need of other components or members like additional screws or bolts. Due to the co-linearity of the axes of the hinge pin and the engaging portion, the position of the pin is ensured.

According to one embodiment of the mobile cooling box, the engaging portion may be a bolt portion, the engaging portion having a male thread. Thereby, mounting is easy and the connection is strong.

According to another embodiment of the mobile cooling box, the pin module further comprises a backing plate between the hinge pin and the engaging portion. The backing plate is positioned in a plane perpendicular to the longitudinal axis of the hinge pin. The backing plate has a pin-side surface and a bolt-side surface. The backing plate has several advantages. It assists at the mounting of the pin module and provides for a stable position thereof ensuring that the pin withstands the forces occurring during movement of the lid while opening or closing the lid.

According to a further embodiment of the mobile cooling box, the backing plate may have a circular shape. Since the complete pin module is screwed or bolted in, the circular shape of the backing plate provides for rotational symmetry. Advantageously, this makes it irrelevant at which rotational position the hinge module is mounted.

According to another embodiment of the mobile cooling box, the pin module may be mounted to the box main body at a vertical surface thereof.

According to a further embodiment of the mobile cooling box, the backing plate abuts with its bolt-side surface against said vertical surface of the box main body. This reliably hinders the pin module from inclining away from the position of the initial longitudinal axis over time during use and thus, contributes to the force resistance of the hinge module.

According to another embodiment of the mobile cooling box, the vertical surface of the box main body has a pin module attachment portion. The pin module attachment

3

portion comprises a bore in which the engaging portion is fastened. Further, the pin module attachment portion comprises a recess for accommodating the backing plate. The recess even more ensures the position of the pin module and hinders inclination of the hinge pin during extensive use. The recess preferably has the same circular shape as the backing plate and, by screwing the pin module in, the backing plate fits with and sinks into the recess.

According to a further embodiment of the mobile cooling box, the recess has a depth corresponding to the thickness of the backing plate. Thus, the transition from said vertical surface of the box main body to the surface of the pin-side surface of the backing plate is essentially flush. This provides for a smooth surface and fewer corners and edges that could be disturbing and contributes to the smooth and robust overall design of the mobile cooling box according to the present embodiments.

According to another embodiment of the mobile cooling box, the hinge pin has a tool engagement portion at its front end for fastening the pin module. This further eases strong fixation of the pin module. The kind of tool engagement portion is not particularly limited. It can be, for example, an outer contour at the hinge pin's front end in the form of a hexagon screw head. However, other forms of tool engagement portions are also possible within the scope of the present embodiments.

According to a further embodiment of the mobile cooling box, the tool engagement portion is a hexagonal socket at the front end face of the hinge pin and the entire hinge pin has a smooth outer surface of a cylindrical shape. Apart from being optically more esthetic, since the entire outer surface of the hinge pin is a smooth cylinder, the hinge construction and the respective pivotal guiding of the lid is more resilient.

According to one embodiment of the mobile cooling box, the entire pin module including the hinge pin, the backing plate and the engaging portion may be formed of metal as one single and integral component. This makes the pin module more robust and contributes to the smooth overall design of the mobile cooling box.

According to another embodiment of the mobile cooling box, the bearing module is present at the lid and the pin module is present at the box main body. Furthermore, the hinge bearing only partly envelops the hinge pin and is open in a direction perpendicular to the longitudinal axis of the hinge pin. Thus, the bearing module allows the hinge pin to be released from the hinge bearing, thereby enabling the lid to be removed completely from the box main body. Removing the lid is of advantage in a number of situations, for example, for cleaning purposes. With one side open of the otherwise pin-surrounding or pin-enveloping hinge bearing, the lid can simply be pulled away from the pivot axis, without the need of any further actions, possibly with only one hand.

According to another embodiment of the mobile cooling box, the bearing module may be configured so that the lid cannot be removed from the box main body when the mobile cooling box is closed. Thus, removing the lid is only possible when the box is open which prevents unintentional lid removal and warming of the inside of the mobile cooling box.

According to a further embodiment of the mobile cooling box, the bearing module may further comprise a spring element configured to hold the hinge pin in place and to provide a predefined and limited resistance when removing the lid from the box main body. Thus, a slight additional force has to be exerted for removing the lid against said

4

resistance. This further prevents the lid from unintentionally falling off the box main body, when the box is open.

According to another embodiment of the mobile cooling box, the bearing module further comprises an abutting portion. The abutting portion, when the lid is pivoted open for a predetermined angle of more than 90°, abuts against a region of the box main body, thereby enabling the lid to rest in the open position. Thus, the lid can stand in a more or less upright position without being held by the user, who, thus, has his or her hands free for loading or unloading the mobile cooling box. Generally, in this embodiment the bearing module with its abutting portion is configured so that the lid can stay in an open position with an angular deflection from the closed position in the range of 91° to 180°, more specifically 91° to 150° and even more specifically 91° to 120°.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. All of the above outlined features are to be understood as exemplary only and many more features and objectives of the various embodiments may be gleaned from the disclosure herein. Therefore, no limiting interpretation of this summary is to be understood without further reading of the entire specification, claims and drawings, included herewith. A more extensive presentation of features, details, utilities, and advantages of the present invention is provided in the following written description of various embodiments of the invention, illustrated in the accompanying drawings, and defined in the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

In the following, embodiments of the mobile cooling box are described in more detail with reference to the accompanying drawings, wherein

FIG. 1 shows a front perspective view of a mobile cooling box according to present embodiments;

FIG. 2 shows a back-perspective view of the mobile cooling box of FIG. 1;

FIG. 3 shows the open mobile cooling box of FIG. 1;

FIG. 4 shows a front perspective view of another mobile cooling box according to present embodiments;

FIG. 5 shows the open mobile cooling box of FIG. 4;

FIG. 6 shows a front perspective view of another mobile cooling box according to present embodiments;

FIG. 7 shows the open mobile cooling box of FIG. 6;

FIG. 8 shows an isolated perspective view of a user interface module;

FIG. 9 shows an exploded view of the component of FIG. 8;

FIG. 10 shows another exploded view of the component of FIG. 8;

FIGS. 11 to 13 illustrate a sequence of a mounting procedure;

FIG. 14 shows a section of the mobile cooling box of FIG. 1 with actuated latch handle;

FIG. 15 shows an exploded view of the latch handle of FIG. 14;

FIG. 16 shows an isolated perspective view of the latch handle of FIG. 14;

FIG. 17 illustrates the working principle of the latch handle of FIG. 14;

5

FIG. 18 illustrates the mounting procedure of a handle module;

FIG. 19 shows an exploded view of the handle module of FIG. 18;

FIGS. 20 and 21 show different perspectives of a cut view of the outer side wall;

FIG. 22 shows a section of the mobile cooling box of FIG. 1 with a hinge module;

FIG. 23 shows relevant parts of FIG. 22;

FIG. 24 shows an inside perspective view of the hinge module of FIG. 22;

FIG. 25 shows a section of the mobile cooling box of FIG. 1 with removed lid;

FIG. 26 shows the section of FIG. 25 and illustrates a mounting procedure;

FIG. 27 illustrates the insertion of an ice maker module into the open mobile cooling box of FIG. 4;

FIGS. 28 and 29 show different perspectives of an exploded view of the ice maker module; and

FIG. 30 shows a section of the open mobile cooling box of FIG. 4 with a lamp system.

DETAILED DESCRIPTION

The illustrated mobile cooling boxes 1 in FIGS. 1 to 7 are generally rectangular in shape. Basically, the mobile cooling boxes according to some embodiments have a box main body 2 and one or a plurality of lids, for example two lids 3 for opening the box 1 and providing access to the inside of the box 1. In the present case, access to the inside of the box 1 is possible from above, but is not limited thereto. The front edge of the lid 3 can be pivotally opened. The rear edge is hinged to the box main body 2. At its front and rear edges, the mobile cooling box 1 is rounded, while the side edges are covered and protected by a fender frame 23 that forms part of the box main body 2. The height of the fender frame 23 is equal to the level of the lid 3 when the mobile cooling box 1 is closed. Thus, the lid 3 when being closed sort of sinks or recesses between the two opposite fender frames 23 thus offering a smooth, uniform and robust look of the mobile cooling box.

In this context and within the framework of the present embodiments, but without limitation, all directional terms, like front, rear, back, upper, lower, above, sink, as well as broadness and depth refer to the mobile cooling box 1 standing on the ground as usually intended and from a perspective facing the side of the mobile cooling box 1 where the edge of the lid is pivotable to the above while opening, unless explicitly stated otherwise.

Each illustrated mobile cooling box 1 is of different depth and width. The lid 3 or the lids 3 are to be opened from a side where the fender frame 23 is not located. This is in case of the mobile cooling box 1 of rather small size, as illustrated in FIGS. 1 to 3, the shorter side of the mobile cooling box 1. In case of the two mobile cooling boxes 1 of rather large size, as illustrated in FIGS. 4 and 5 and FIGS. 6 and 7, it is the longer side of the mobile cooling box 1, respectively.

The mobile cooling box 1 has an electrically driven cooling unit and comprises an internal battery (not shown). The mobile cooling box 1 can be used in plugged-in mode or in battery mode. The mobile cooling box 1, therefore, has the required sockets 24 located at one of its sides.

In the following, different aspects and features of the mobile cooling box are described. As will become apparent, many of the following aspects relate to readily mountable

6

modules for various functions which can be mounted without limitation to the mobile cooling box 1 regardless of the size thereof.

The FIGS. 8 to 10 show a user interface. With such a user interface it is possible for the user to sort of communicate with the mobile cooling box 1, that is, retrieving information from the mobile cooling box 1 and entering controls into the mobile cooling box 1. The mobile cooling box 1 is equipped with the user interface module 100 for operation and control by the user. The user interface module 100 is mounted at the mobile cooling box 1 in a way that a part of it is engaged with a designated opening at the mobile cooling box 1 on one side of the user interface module 100 and fixed with additional fixation structure at the other side of the user interface module 100. For example, as illustrated, screws 170 are used for fixing the module 100 at the left side, however other structures may be utilized. On the right side, no screw is necessary. At this side the module 100 is engaged with the designated opening at the mobile cooling box.

As shown in FIGS. 9 and 10, the user interface module 100 is an assembly of components, namely a circuit board 110, a housing 120 and a front cover 130. The assembly is mounted in a recessed part of the mobile cooling box 1 so that essentially only the front cover 130 is directly visible for the user.

The circuit board 110 forms a latch 111 extending away from the right side of the assembly with respect to the housing 120 and the front cover 130. The latch 111 is engaged with the designated opening at the mobile cooling box 1. The circuit board 110 is essentially longer in size than the housing 120 at the right side. The circuit board 110 extends over the edge of the housing 120 and the front cover 130. In the illustrated embodiment, the circuit board 110 extends over the edge of the housing 120 and the front cover 130 for about 1 cm, but also other dimensions are possible. At the backside of the circuit board 110 the part forming the latch 111 is further strengthened by additional material provided in this area.

As shown in FIG. 10, the circuit board 110 has a connector 112. The connector 112 is located at the backside of the circuit board 110. The user interface module 100 is electronically connected with the mobile cooling box 1 by using the connector 112. A respective plug (not shown) is provided in the inside of the box main body 2 of the mobile cooling box 1 and can be reached from the opening in which the user interface module 100 is hooked.

The user interface module 100 further has a USB port 113. In the illustrated embodiment a single USB port 113 is provided. However, there can be also a plurality of USB ports provided, for example depending on the size of the mobile cooling box. The USB port 113 is present at the circuit board 110. Moreover, two through holes for a screw connection are provided. By using the USB port 113 the user can recharge external devices like batteries, lamps, smart-phones, etc. Moreover, the USB port 113 provides access to the internal control of the mobile cooling box 1 and, depending on the settings and version, enables download of internal data and/or programming of functions of the mobile cooling box 1.

The housing 120 is clamped onto or over the circuit board 110 by respective clip-in elements 121. In the shown example, three clip-in elements 121 on each of the upper and the lower side of the housing 120 are sufficient to achieve a stable and robust fixation of the housing 120. The housing 120 is formed of an injection molded plastic component. The housing provides co-injected regions of plastic that is softer than at other regions of the housing 120.

At the housing **120**, the area of the USB port **113** is left open to allow access to the USB port **113**. A rubber cap **140** is provided for covering the USB port **113** when not in use. In the present embodiment, the rubber cap **140** is swingably attached to the housing **120** to avoid losing the rubber cap **140**. By removing the rubber cap **140** from the housing **120** access to the USB port **113** becomes possible. Here, the size and design of the rubber cap **140** ensures coverage of the through holes provided for the screw connection at the same time.

The user interface module **100** further has a display **160**, the display **160** is arranged behind the front cover **130** and the front cover **130** of the user interface module **100** is transparent at least in the area of the display **160**.

The user interface module **100** further has three operation devices, here in the form of buttons **150** extending from the user interface module **100**. One of the buttons **150** is located at the right side of the display **160** and provides an up-and-down selection button **150** for navigating through the menu of the control menu of the implemented software. In the illustrated embodiment, two further buttons **150**, here designed in form of single round buttons, are provided at both sides next to the display **160**, or, respectively, the afore-mentioned button **150**. The buttons **150** are made of rubber, or covered by rubber, in order to provide good haptics and provide a stable and robust design.

The front cover **130** user interface module **100** is of a scratch resistant material or has a scratch resistant coating.

The user interface module **100** is capable of wireless communication with an external electronic device, for example by Bluetooth technology, WLAN or any other suitable technology. The external electronic device can be a remote control, a smartphone or the like. Hence, the user interface **100** and thus the mobile cooling box is remote controllable with the external electronic device. In case of using a smartphone, a respective app is available and to be used on the smartphone.

The user interface module **100** provides to the user functions and controls like ON/OFF-switching the cooling, temperature control including setting, if desired in a time-shift manner, displaying current temperature, temperature history graphs, temperature type setting ($^{\circ}$ C./ $^{\circ}$ F.), alarm setting, energy saving mode, displaying battery status, including voltage level and/or battery remaining time, power consumption history graphs, lid **3** open indication, wireless communication ON/OFF and setting display brightness. The below list shall not be construed as conclusive. Further functions are, thus, also possible.

For mounting the user interface module **100**, the part on the right side of the user interface module **100** that is supposed to be engaged with the designated opening at the mobile cooling box **1** is laterally slid into the designated opening. This step is illustrated in FIG. **11**. In FIG. **11** the rubber cap **140** is shown in its open position. However, the rubber cap **140** itself is not involved in the inserting and assembling process of the user interface module **100**. After being inserted with the latch **111**, the left side of the user interface module **100** is rotated in place as shown in FIG. **12** and, in the next step, the user interface module **100** is fixed with two screws to complete the mounting process.

The mobile cooling box **1** according to some embodiments and as described before has at least one lid **3**. By using the lid **3** the mobile cooling box **1** can be opened from one side-edge of the lid **3**. Thereby, access is provided to the inside of the box **1**. At the opposite side-edge of the lid **3**, the lid **3** is hinged to the box main body **2**. By this hinge connection the lid **3** can be pivoted upwards.

As shown in FIGS. **14** to **17**, the mobile cooling box **11** is equipped with a latch handle module **200**. The latch handle module **200** allows manually locking and unlocking of the lid **3** and, thus, opening and closing the mobile cooling box **1** by the lid **3**. The latch handle module **200** is integrated in the lid **3** and located at the front side edge of the lid **3** of the illustrated embodiment of the mobile cooling box **1**.

The latch handle module **200** is an assembly of components including an actuating element **201**, a locking element **202** and a casing **203**. The actuating element **201** is manually operable by the user. The locking element **202** is engageable with a corresponding counterpart at the box main body **2**. By engaging the corresponding counterpart at the box main body **2** the lid **3** is locked from being opened.

The latch handle module **200** provides a mechanism for locking and unlocking the lid **3**. According to the latch handle module **200** the actuating element **201** and the locking element **202** are mechanically connected to each other. As illustrated in FIG. **17**, by operating the actuating element **201**, the locking element **202** disengages with its corresponding counterpart at the box main body **2** and allows up-folding the lid **3**. In this embodiment, the locking element **202** is designed as a snap-in latch. The snap-in latch, in a closed state, extends into the corresponding counterpart being a recess in the box main body **2**. Furthermore, the actuating element **201** has a surface which can be pushed by the fingers of the user's hand. In the illustrated embodiment the actuating element **201** has a width of about 10 cm. However, according to the overall size the width of the actuating element **201** can have also a different size. For opening the lid **3**, the actuating element **201** is pivoted about an axis of rotation with a pivoting direction that is the same as that of the lid **3** when being opened. Therefore, there are no opposing movements for the user's hand, which has been found to be comfortable for the user.

As regards the working principle of the latch handle module **200**, the latch handle module **200** further comprises a shaft **204**. The shaft **204** has a longitudinal axis being co-linear with the axis of rotation of the actuating element **201**. The actuating element **201** is connected to and pivotable about the shaft **204**. The shaft **204** is of a rigid metal material and extends essentially over the entire width of the latch handle module **200**. The latch handle module **200** further has two springs **205** by means of which the mechanism provided by the latch handle module **200** is spring loaded. The mechanism provided by the latch handle module **200** is spring loaded for providing a restoring force that ensures that the actuating element **201** and the locking element **202** return to their respective initial positions after an operation of the actuating element **201** by the user.

As shown in FIGS. **18** and **19**, the mobile cooling box **1** is equipped with two handle modules **300**. The two handle modules **300** are located at an outer side surface of the box main body **2**. One handle module **300** has a handlebar **301**. The handlebar **301** is intended to be grasped by the hand of the user and has a longitudinal axis as well as two ends, two hangers **302** and two brackets **303**. The handlebar **301** is attached at its two ends to the two hangers **302**. The hangers **302** are rotatably mounted at the two brackets **303**. The two brackets are fixed to the outer side surface of the box main body **2**.

The handle module **300** is designed in a way that the handlebar **301** hangs downwards in an unactuated state and can be swung out and upwards for carrying the mobile cooling box **1**.

Each of the brackets **303** comprises a mounting area, or mount, **304** and a shielding area, or shield, **305**. The mount-

ing area **304** faces the outer side surface of the box main body **2** to which the bracket **303** is fixed. The shielding area **305** hides the hangers **302** and the handlebar **301** in an unactuated state of the handle module **300** and in a lateral perspective along the longitudinal axis of the handlebar **301**.

The handle module **300** is designed so that, in an unactuated state of the handle module **300** and in a lateral perspective along the longitudinal axis of the handlebar **301**, at least a section of the outer contour of the shielding area **305** is flush with the handlebar **301** and with the hangers **302**. Thus, when the mobile cooling box **1** is not carried, the handlebar **301** with its hangers **302** exactly hides behind the bracket **303** in the respective lateral perspective.

The handlebar **301** and its hangers **302** are spring-loaded. Thus, in an unactuated state, the handlebar **301** and the hangers **302** are forced in a direction to the mobile cooling box **1** and are thus kept hidden in-between the shielding areas **305** of both brackets **303**. For this purpose, two springs **308** are arranged within the handle module **300**. The springs **308** force the hangers **302** relative to the brackets **303** to abut against the part with the mounting area **304**.

The handle module **300** is designed in a way that, in an actuated state, the hangers **302** with the handlebar **301** are swung out and upwards and rest in a position relative to the mobile cooling box **1**. Thus, the mobile cooling box **1** can be carried in a comfortable way. The hangers **302** with the handlebar **301** rest in the position by means of a region of the hangers **302** abutting against a region of the brackets **303**. Thereby, at the joint between the brackets **303** and the hangers **302**, the hangers are rounded in a section around the respective pivot axis. Moreover, a corresponding roundness is present at the brackets **303** to the extent that, when the hangers pivot out, the round part of the brackets **303** that enclose the round part of the hangers abut against the flanks of the hangers **302**. Thus, further rotation of the hangers **302** is blocked.

Furthermore, at its mounting area **304** each bracket **303** comprises two through holes **306** for fixing the bracket **303** to the outer side surface of the box main body **2** by means of fixing elements **307**. In the illustrated embodiment of the mobile cooling box the fixing elements are designed in the form of screws but are not limited thereto. The through holes **306** and the respective fixing elements **307** are covered by the hanger **302** that is mounted to said bracket **303**, in an unactuated state of the handle module **300**. Thereby, the hanger **302** abuts against said mounting area **304**.

An additional accessory, like for example a bottle opener (not shown) or other equipment or tooling, can be attached at the through holes **306** by respective means, like for example screws **309**.

As mentioned, the mobile cooling box **1** is basically rectangular in shape and has different dimensions in width and depth and height. Further, the two handle modules **300** are located at the respective two shorter outer side surfaces of the mobile cooling box **1** being opposite to each other. Thereby, when carrying the mobile cooling box **1** a tilting of the mobile cooling box **1** can be avoided.

In the present embodiment the handlebar **301** has a circular cross-section. Moreover, the handlebar **301** has a length of at least 10 cm to ease gripping the handlebar by the user's hand. However, other dimensions are also possible. The lower part of the hangers **302** correspond with this rounded contour. Also, the lower part of the brackets **303** partly correspond with this contour. Hence, the components are flush in an unactuated state.

At least the handlebar **301**, the hangers **302** and the brackets **303** of the handle module **300** are made of alumi-

num. At least part of the surface of the aluminum is roughened and has an oxidic protective layer.

As illustrated in FIGS. **20** and **21**, the mobile cooling box **1** is equipped with air vents **400**. The air vents **400** are located on at least one side wall of the box main body **2**. In the periphery of the air vents **400** cord fixation means are present (not shown). The electrical cord (not shown) provided for connecting the mobile cooling box to electrical power can be, especially in case the cord is not in use, attached to the outside of the mobile cooling box **1** in a known manner. The cord fixation means can for example be formed in the shape of hooks to which the cord can be removably attached. Nearby the air vents **400**, as illustrated in FIG. **20**, at least one power connector is present to connect the removable power cord (not shown) to the mobile cooling box **1** to supply electrical power to the mobile cooling box **1**.

The air vents **400** comprise a plurality of horizontal opening or slots **401** (in the following generally referred to as slots), respectively, allowing air circulation through the respective side wall of the mobile cooling box **1**. The slots **401** comprise shielding elements **402** protruding inside the mobile cooling box **1**. Each of the shielding elements **402** is designed in such a way that the shielding element at least partly blocks the view into the inside of the mobile cooling box **1** from the outside. In other words, the inside of the mobile cooling box **1** is not visible from the outside due to the design of the shielding elements **402**.

One slot **401** has an upper edge **403** and a lower edge **404**. Both, the upper edge **403** and the lower edge **404** lie in the plane of the respective side wall. One of the shielding elements **402** extends from the lower edge **404** to the inside of the mobile cooling box **1** and further upwards with respect to said lower edge **404**, virtually in the direction of and at least up to the height of the upper edge **403**. Thus, the inside of the mobile cooling box **1** is not visible from the outside due to the design of the shielding element **402**.

Particularly, in a vertical cross section perpendicular to said side wall, the shielding element **402** extends from the lower edge **404** in upward curved form, namely in the form of a segment of a circle.

Furthermore, one of the shielding elements **402** extends from the upper edge **403** to the inside of the mobile cooling box **1**.

Particularly, in a vertical cross section perpendicular to said side wall, the shielding element **402** extends from the upper edge **403** to the inside of the mobile cooling box **1** in a straight horizontal direction. This has essentially the function of providing more stability to the side wall and to uniform the upper and lower edges **403** and **404** with regard to the roundness.

The side wall where the vents are present is manufactured together with the shielding elements **402** as a one-piece component which is made of plastic and manufactured by injection molding.

As is shown in FIGS. **22-26**, the mobile cooling box **1** has a lid **3**. The lid **3** is pivotally attached to the box main body **2** by means of two hinge modules **500**. Each hinge module **500** comprises a pin module **510**. The pin module **510** has a hinge pin **511** with a front end, a rear end, a longitudinal axis about which the lid **3** is pivotable, and a smooth outer surface having a cylindrical shape.

The hinge module **500** further comprises a bearing module **530**. The bearing module **530** has a hinge bearing **531** accommodating the hinge pin **511**. The hinge pin **511** laterally extends with its front end into the hinge bearing **531**. Thus, during pivoting the lid **3** with respect to the box

11

main body 2 an axis of the hinge bearing 531 remains co-linear with the longitudinal axis of the hinge pin 511.

For the mounting of the pin module 510, the pin module 510 further comprises an engaging portion 513, here in form of a bolt portion 513. The bolt portion 513 has a male thread and extends from the rear end of the hinge pin 511. The bolt portion 513 has a longitudinal axis being co-linear to that of the hinge pin 511.

The pin module 510 further comprises a backing plate 514 between the hinge pin 511 and the bolt portion 513. The backing plate 514 lies in a plane perpendicular to the longitudinal axis of the hinge pin 511 and has a pin-side surface and a bolt-side surface. The backing plate 514 has a circular shape so that it is symmetrical with regard to rotation.

The pin module 510 is mounted to the box main body 2 at a vertical surface thereof which is the inner sider of a part of the box main body 2. The backing plate 514 abuts with its bolt-side surface against said vertical surface of the box main body 2.

Furthermore, the vertical surface of the box main body 2 to which the pin module 510 is attached to has a pin module attachment portion 520. The pin module attachment portion 520 comprises a bore 521 having a female thread, in which the bolt portion 513 is fastened, and a recess 522 for accommodating the backing plate 514. The recess 522 has a depth corresponding to the thickness of the backing plate 514. Hence, the transition from said vertical surface of the box main body 2 to the surface of the pin-side surface of the backing plate 514 is flush. In order to provide for sufficient stability, the thickness of the backing plate 514 is about 2 mm.

Furthermore, the hinge pin 511 has a tool engagement portion 512 at its front end for fastening the pin module 510. The tool engagement portion 512 is a hexagonal socket that is engageable with a hex key at the front end face of the hinge pin 511. Moreover, the entire hinge pin 511 has a smooth outer surface of a cylindrical shape, so that the pivoting movement can be guided over the entire length of the hinge pin 511.

The entire pin module 510 including the hinge pin 511, the backing plate 514 and the bolt portion 513 is formed of metal. Moreover, the entire pin module 510 is formed as one single and integral component. Thus, the pin module 510 is very robust component.

The bearing module 530 is present at the lid 3 and the pin module 510 is present at the box main body 2. The hinge bearing 531 only partly envelops the hinge pin 511 and is open in a direction perpendicular to the longitudinal axis of the hinge pin 511. Thus, the bearing module 530 allows the hinge pin 511 to be released from the hinge bearing 531, thereby enabling the lid 3 to be removed completely from the box main body 2. In particular, when it is pivoted in an open direction for about 60° and more the lid 3 can be removed. Thus, the bearing module 530 is configured so that the lid 3 cannot be removed from the box main body 2 when the mobile cooling box 1 is closed.

The bearing module 530 further comprises a spring element 533. The spring element 533 protrudes out of an upper surface part of the hinge bearing 531. The spring element 533 is configured to hold the hinge pin 511 within the hinge bearing 531 and to provide a certain resistance during removing the lid 3 from the box main body 2.

The bearing module 530 further comprises an abutting portion 532. When the lid 3 is pivoted in the open direction for an angle of about 100° the abutting portion 532 abuts

12

against a region of the box main body 2. Thereby, the lid 3 is enabled to rest in an open position.

As shown in FIGS. 27 to 29 the mobile cooling box 1 has an inside that is laminated with a lining 601 at the inner side walls 21 and at the floor 22 of the box main body 2.

The mobile cooling box 1 is equipped with an ice maker module 600. The ice maker module 600 has a freezing compartment 606. The ice maker module 600 can be removably placed on a freezing zone 602 on a floor part of the lining 601.

The mobile cooling box 1 further comprises an evaporator 603 arranged underneath the lining 601 at the freezing zone 602, for providing sufficient cooling power for freezing goods.

The ice maker module 600 is an assembly of components, namely a frame 604 and a cover 607. The frame 604 has lateral walls 605 limiting the freezing compartment 606. The cover 607 is attached to the upper side of the frame 604 for opening and closing the ice maker module 600 and providing access from above to the freezing compartment 606. The freezing compartment 606 is limited at its ground by the lining 601 at the freezing zone 602. Thus, the goods to freeze are placed directly on the floor part of the freezing zone 602 for efficient freezing.

The freezing zone 602 is rectangular and is located in a niche limited by the lining 601 of three of the inner side walls 21. The ice maker module 600 fits in the niche.

At least one pair of corresponding attachment means 608 configured to releasably engage with each other is present at the lining 601 of the inner side walls 21 adjacent to the freezing zone 602 and at the ice maker module 600, respectively. By the at least one pair of corresponding attachment means 608 the position of the ice maker module 600 is secured. The pair of attachment means 608 provides for a form-locked connection being a snap-in connection. The snap-in connection consists of hook and a corresponding recess. The hook is a projecting element that is configured to snap in the recess. The hook is located at the ice maker module 600 and the corresponding recess is located at the lining 601 of the respective inner side wall 21. The hook is located at the frame 604 of the ice maker module 600.

The hook and the recess of one pair of corresponding attachment means 608 are formed as integral parts of the lining 601 and the ice maker module 600, respectively.

Furthermore, the cover 607 is hinged to the frame 604. Thus, the cover 607 is swingably openable to the above and can be opened about an angle of about 100°. The cover 607 has a grip portion 609 by means of which the cover 607 can be opened and closed by the hand of the user.

The ice maker module 600 further comprises two ice trays 610. The ice trays 610 fit into the freezing compartment 606. Each of the ice trays 610 is equipped with a cap 611. Each ice tray 610 has a plurality of recesses for forming ice cubes. The cap 611 has small holes 612 in form of bores with a rather small diameter. By these holes air exchange is enabled between inside and outside of the ice tray, but predominantly preventing water from leaking out.

The opening of the tiny holes has a cross section of about 0.20 mm. Above each recess, one of the tiny holes is arranged.

As shown in FIG. 30 the mobile cooling box 1 comprises a lamp module 700. By the lamp module 700 light can be provided in the inside of the box main body 2. To turn on the lamp module 700 it does not have any mechanical switches as in usual refrigerators. In the present embodiment of the lamp module 700 can be switched ON or OFF by means of a reed sensor (not shown). The front cover of the lamp

13

module **700** is perfectly flush with the surface at which the lamp module **700** is arranged.

The front cover is mounted to the inner lining in a waterproof manner. Specifically, the front cover of the lamp module **700** is clipped in a corresponding recessed part of the inner lining and is equipped with sealed portions.

Furthermore, the front cover of the lamp module **700** is transparent and provides a diffuse light. The light is emitted from diodes inside the lamp module **700** and both, the light-emitting diodes and the reed sensor are mounted on a circuit board of the lamp module **700**.

For switching the light ON and OFF, a magnet is incorporated in the part of the lid **3** that functionally corresponds with the reed sensor. In the closed state of the lid **3**, the magnet is located in the vicinity of the light module **700** so that the light module is switch OFF. While opening or in the opened state the distance of the magnet, thus, is increased and the light module is switch ON by the reed sensor.

REFERENCE SIGNS

1 Mobile cooling box
2 Box main body
3 Lid
21 Inner side wall of box main body
22 Floor of box main body
23 Fender frame
24 Socket
25 Outer side wall of box main body
100 User interface
110 Circuit board
111 Latch
112 Connector
113 USB port
120 Housing
121 Clip-in element
130 Front cover
140 Rubber cap
150 Operation device/button
160 Display
170 Screw
200 Latch handle module
201 Actuating element
202 Locking element
203 Casing
204 Shaft
205 Spring at the latch handle
300 Handle module
301 Handlebar
302 Hanger
303 Bracket
304 Mounting area
305 Shielding area
306 Through hole
307 Fixing elements
308 Springs at the handle
309 Screws of the handle
400 Air vents
401 Opening/slot
402 Shielding element
403 Upper edge
404 Lower edge
500 Hinge module
510 Pin module
511 Hinge pin
512 Tool engagement portion
513 Engaging portion/bolt portion

14

514 Backing plate
520 Pin module attachment portion
521 Bore
522 Recess
530 Bearing module
531 Hinge bearing
532 Abutting portion
533 Spring element
600 Ice maker module
601 Lining
602 Freezing zone
603 Evaporator for the freezing zone
604 Frame
605 Walls of frame
606 Freezing compartment
607 Cover of ice maker module
608 Attachment means
609 grip portion
610 Ice tray
611 Ice tray cap
612 Tiny holes
700 Lamp module

The invention claimed is:

1. A mobile cooling box, comprising:
 - a box main body, having a first pair of parallel walls and a second pair of parallel walls extending from a bottom of said box main body;
 - upper ends of said walls defining an opening wherein a lid is positioned, said lid pivotable between an opened position and a closed position to access an interior of the box main body;
 - a first hinge module and a second hinge module disposed between said lid and said box main body;
 - each of said first hinge module and said second hinge module having a hinge pin which is fixed relative to said box main body and engages said lid when said lid is in said closed position;
 - a backing plate, said hinge pin extending from said backing plate;
 - an engaging portion extending opposite from said hinge pin, said engaging portion extending into a wall of said box main body;
 - each said backing plate positioned against one wall of said walls;
 - a hinge bearing which engages said hinge pin, said hinge bearing partly enveloping said hinge pin so that said hinge pin may be selectively released from said hinge bearing.
2. The mobile cooling box of claim 1, said engaging portion comprising a thread.
3. The mobile cooling box of claim 1, said engaging portion defined by a bolt.
4. The mobile cooling box of claim 1, said engaging portion being co-linear to said hinge pin.
5. The mobile cooling box of claim 1 the hinge pin having a smooth outer surface.
6. The mobile cooling box of claim 1, wherein said hinge bearing is capable of being selectively released from said hinge pin when said lid is opened a preselected amount or to a preselected position.
7. The mobile cooling box of claim 1, said first pair of parallel walls being longer than said second pair of parallel walls.
8. The mobile cooling box of claim 1, said lid pivoting along one wall of said longer walls or one wall of said shorter walls.

15

9. The mobile cooling box of claim 1, said hinge bearing having an opening at which said hinge pin may be released.

10. The mobile cooling box of claim 9, further comprising a retention structure extending into said opening.

11. The mobile cooling box of claim 10, said retention structure being biased to one position. 5

12. The mobile cooling box of claim 11, said retention structure being movable.

13. The mobile cooling box of claim 12, said retention structure being a spring. 10

14. The mobile cooling box of claim 1, said lid being a first lid and a second lid.

15. The mobile cooling box of claim 1, said walls being two long walls and two short walls.

16. The mobile cooling box of claim 14 further comprising a floor having a first elevation and a second elevation. 15

17. The mobile cooling box of claim 16, said second elevation being higher than said first floor elevation, wherein cooling components are disposed beneath said floor at said second elevation. 20

18. The mobile cooling box of claim 17, further comprising a lining disposed along an interior of said box main body.

* * * * *

16